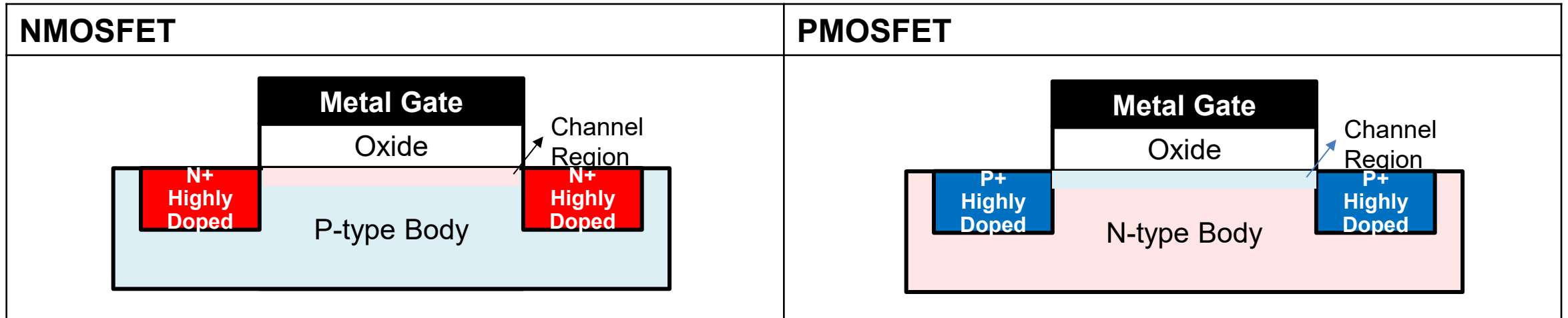


Background-1: V_{th} & Work Function

What is Threshold Voltage (V_{th})?

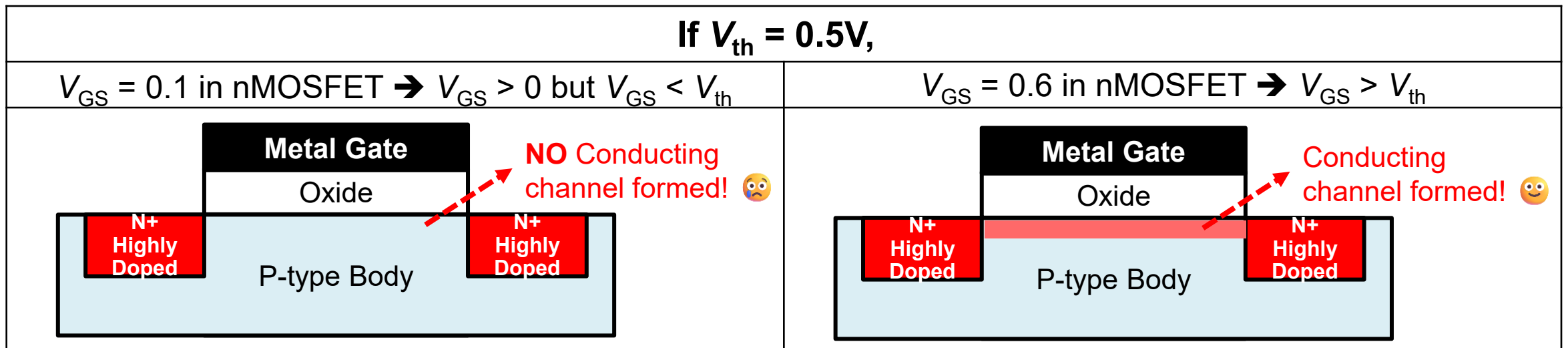
- MOSFET turns on when the gate can form a conducting channel.
- Of the two highly doped regions (shown below figure), the side that supplies the carriers is biased at the less carrier-attractive potential. → This terminal is called the **source** according to its role.
- Therefore, the gate must then be biased sufficiently attractive to the carriers *relative to the source*. For an nMOS, this means applying a sufficiently positive VGS. For a pMOS sufficiently negative VGS is required.
- For some reason (see next slide), there exists the minimum V_{GS} required to form a conducting channel, which is called the **threshold voltage, V_{th}** .
- Therefore, the voltage that actually drives the transistor beyond threshold is $V_{GS} - V_{th}$



Why Does V_{th} Exist?

→ Different Energy Zero Reference for Gate/Silicon

- Why doesn't an nMOS turn on as soon as V_{GS} becomes slightly positive? (Left-bottom Figure) That is, if electrons are attracted to a higher voltage, shouldn't any $V_{GS} > 0$ immediately create a channel?
- The gate and silicon are different materials. **Electrons in the two materials do not start from the same energy reference.** In other words, the gate and silicon have *their own "energy zero point"* — their electron-energy worlds are already offset from each other.
- Therefore, an externally applied V_{GS} does not directly translate into the same amount of attraction at the silicon surface.
- Part of the gate voltage must first overcome this initial energy/potential offset before a strong conducting channel can form, which is the form of V_{th} .



Where Does the Energy Reference Difference Come From?

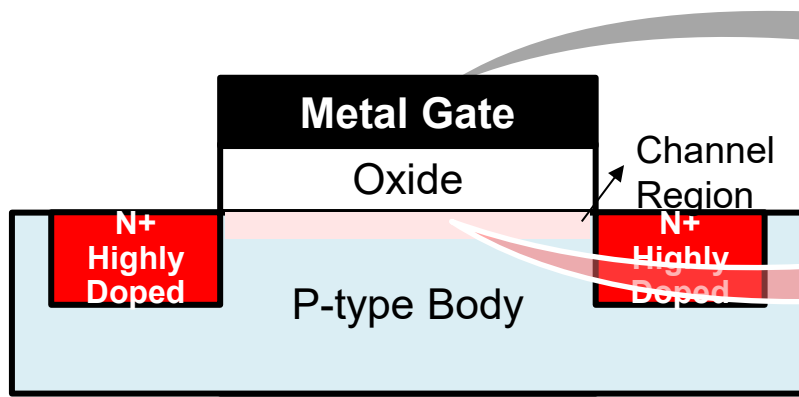
- Different materials interact with electrons differently. Therefore, electrons do not have the same energy reference in every material.
- One of the main factors describing this material-dependent energy reference is the work function, Φ . **Work function (Φ) represent show strongly a material holds its electrons relative to the vacuum level.**
 - More precisely, it is the energy required to move an electron from the material's electron-energy reference (Fermi level) to the vacuum level.
- The gate metal and the silicon therefore have their own work functions:

Φ_M : Metal Work Function

Φ_S : Semiconductor Work Function

- If $\Phi_M \neq \Phi_S$, the gate and silicon start with different electron-energy references.

➔ **This work-function difference creates an initial energy/potential offset and therefore affects the gate-to-source voltage required to form a channel, yielding non-zero V_{th} voltage.**



Different **Gate** and **Body** Conditions

→ Different **Work Function** Difference $\Phi_M - \Phi_S$

→ Different V_{th}

Gate Metal — Φ_M

Different metals have different work functions.

Gate Metal	Φ_M [eV]
Al	4.10
Ta	4.25
TaN	4.45
W	4.60
TiN	4.65
Pt	5.30

Body Region — Φ_S

Even for the same Si, different body doping gives different work functions.

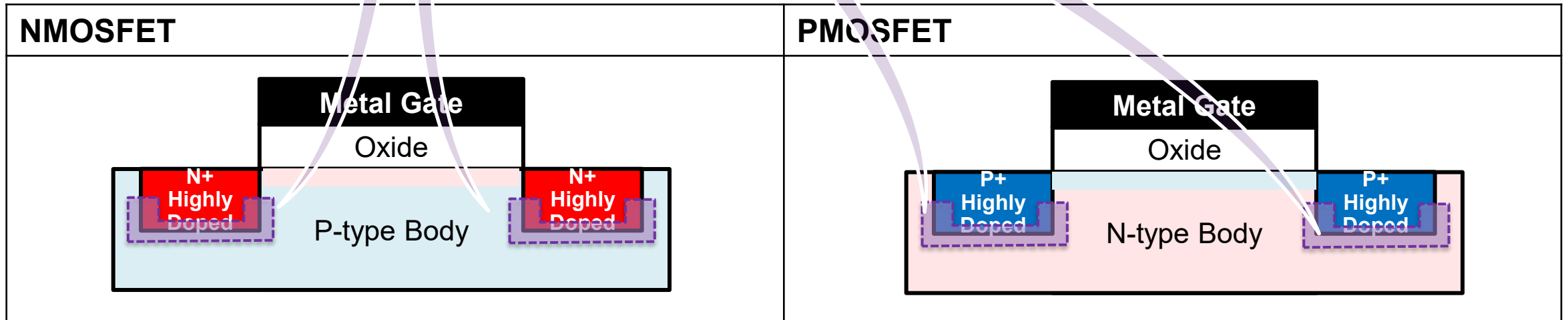
Body Doping	Φ_S [eV]
$p, N_A = 10^{15} \text{ cm}^{-3}$	~4.91
$p, N_A = 10^{16} \text{ cm}^{-3}$	~4.97
$p, N_A = 10^{17} \text{ cm}^{-3}$	~5.03

- Both Φ_M and Φ_S can be changed by device design.
- Therefore, gate material and body doping can both be used to tune V_{th} .

Background-2: Body Contact, Ohmic/Shot

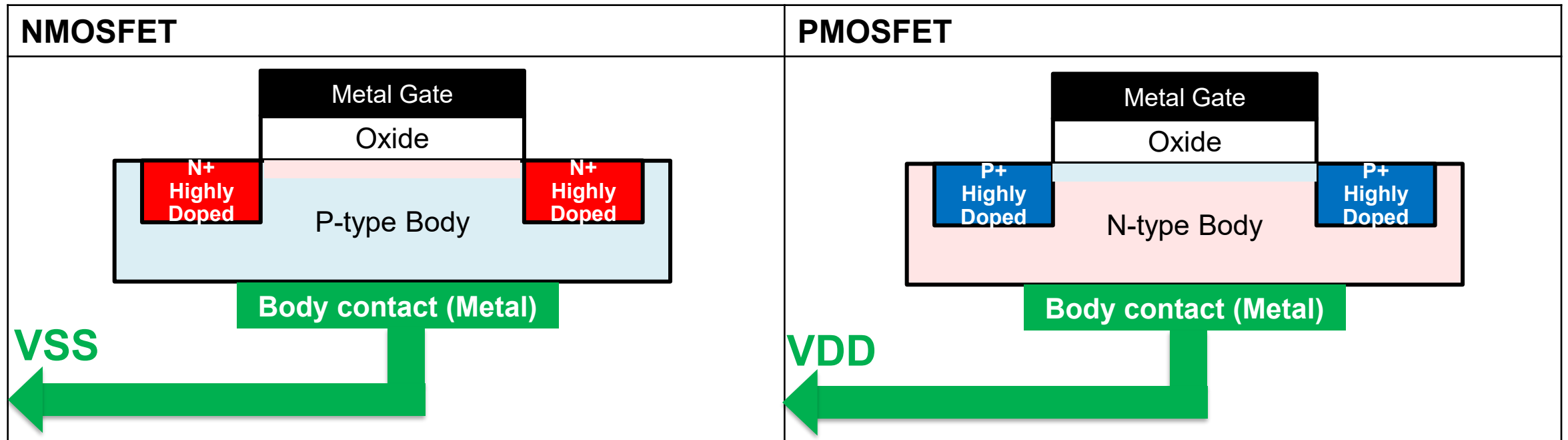
Why Do We Need a Body Contact?

- The source and drain are highly doped regions formed inside the body.
- Therefore, **source/body and drain/body junctions** are always present.
- These junctions should not accidentally conduct current during normal operation.
- The body voltage must therefore be kept at a well-defined potential.
 - Body voltage are required for **obtaining reverse bias P-N junction** for source/body and drain/body junctions
(You remember? You learned in the lecture, **PMOS n-type body: VDD, NMOS p-type body VSS**)



How Can We Make Connect VDD or VSS for Body?

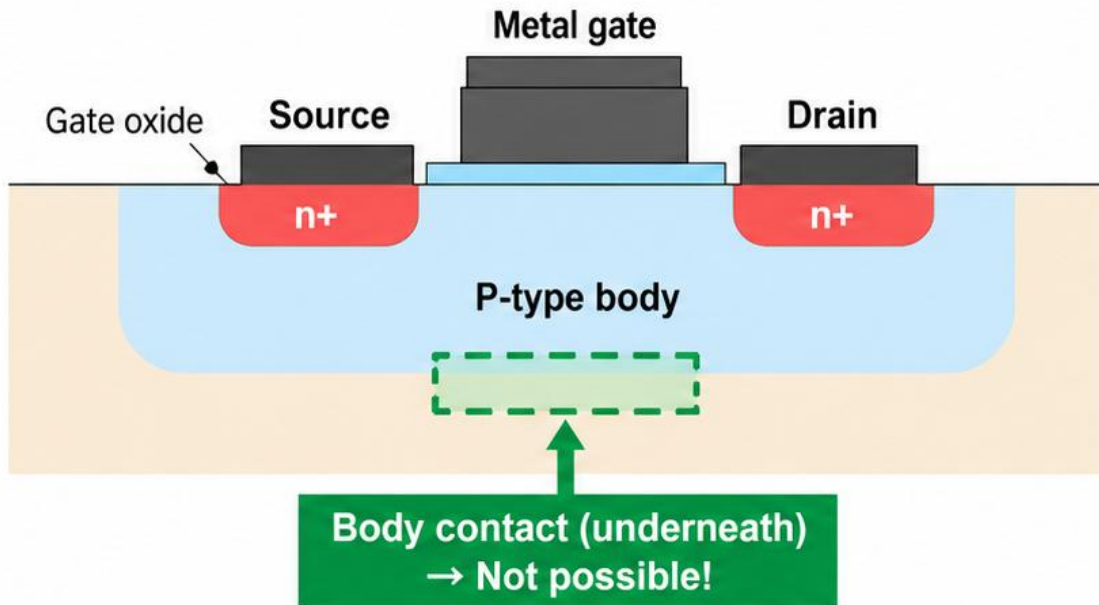
- Okay, “**PMOS n-type body: VDD, NMOS p-type body VSS**” is required for reverse biasing P-N junctions in between S/D and body.
- But how can we connect such voltages to Body? Like “**Body contact**” shown below?



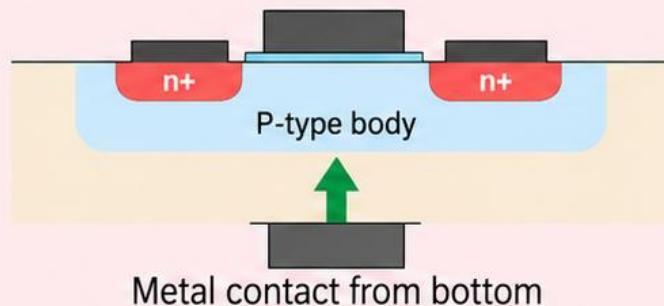
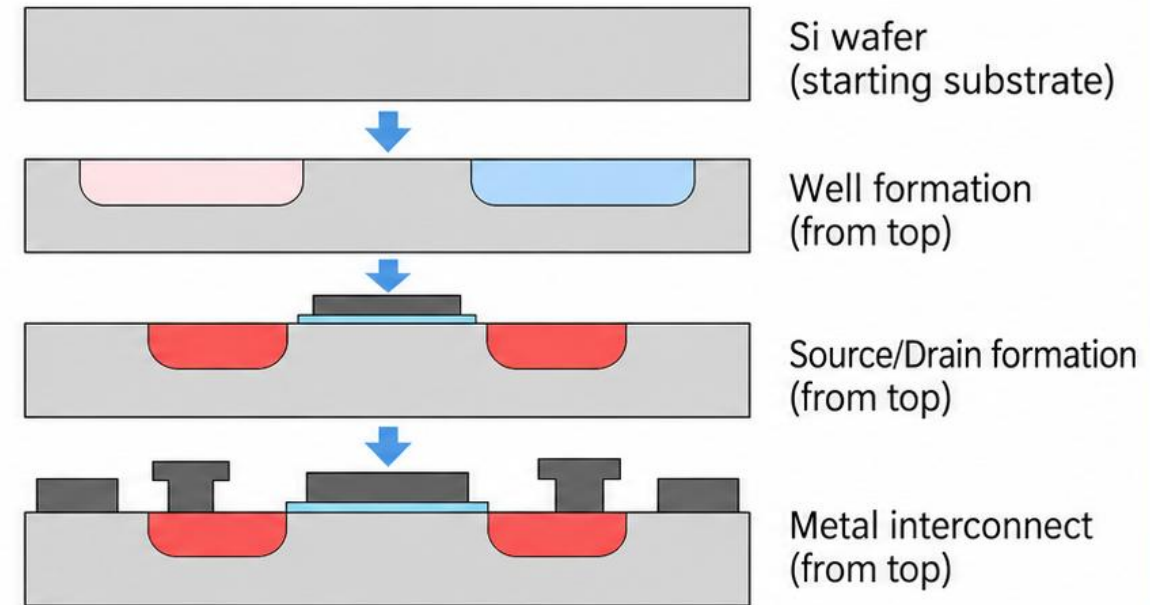
- No! it is not possible! This is because CMOS devices are fabricated from the top surface, while the body region is buried inside the wafer, a direct bottom body contact is not feasible. (See next slide)

CMOS is Fabricated from Top Surface Only

(a) Cross-section of a finished nMOS



(b) CMOS fabrication process (simplified)

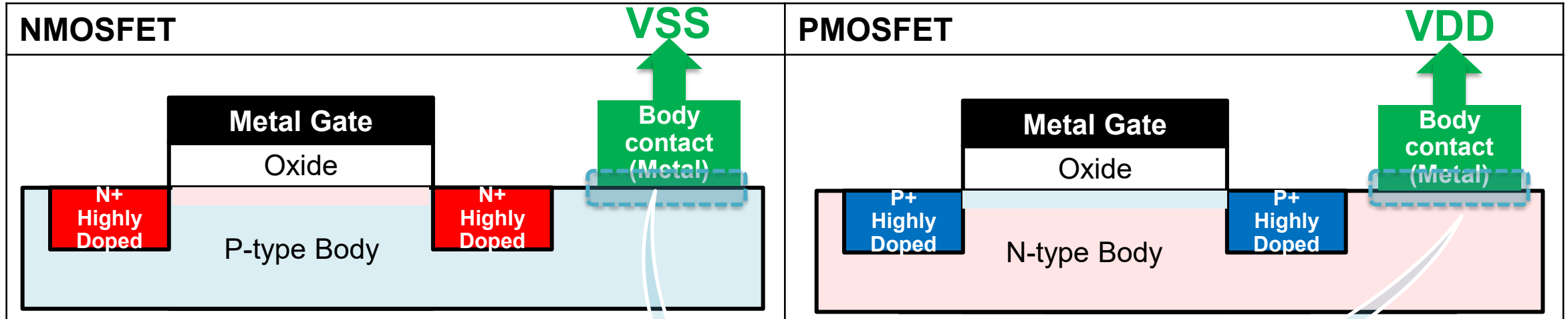


Not possible in standard CMOS process

- All process steps and metal contacts are formed from the top surface of the wafer.
- The body is located inside the silicon wafer (buried).
- Therefore, we cannot place a metal contact underneath the body in a standard top-down CMOS process.

Then, Where Should the Body Be Connected?

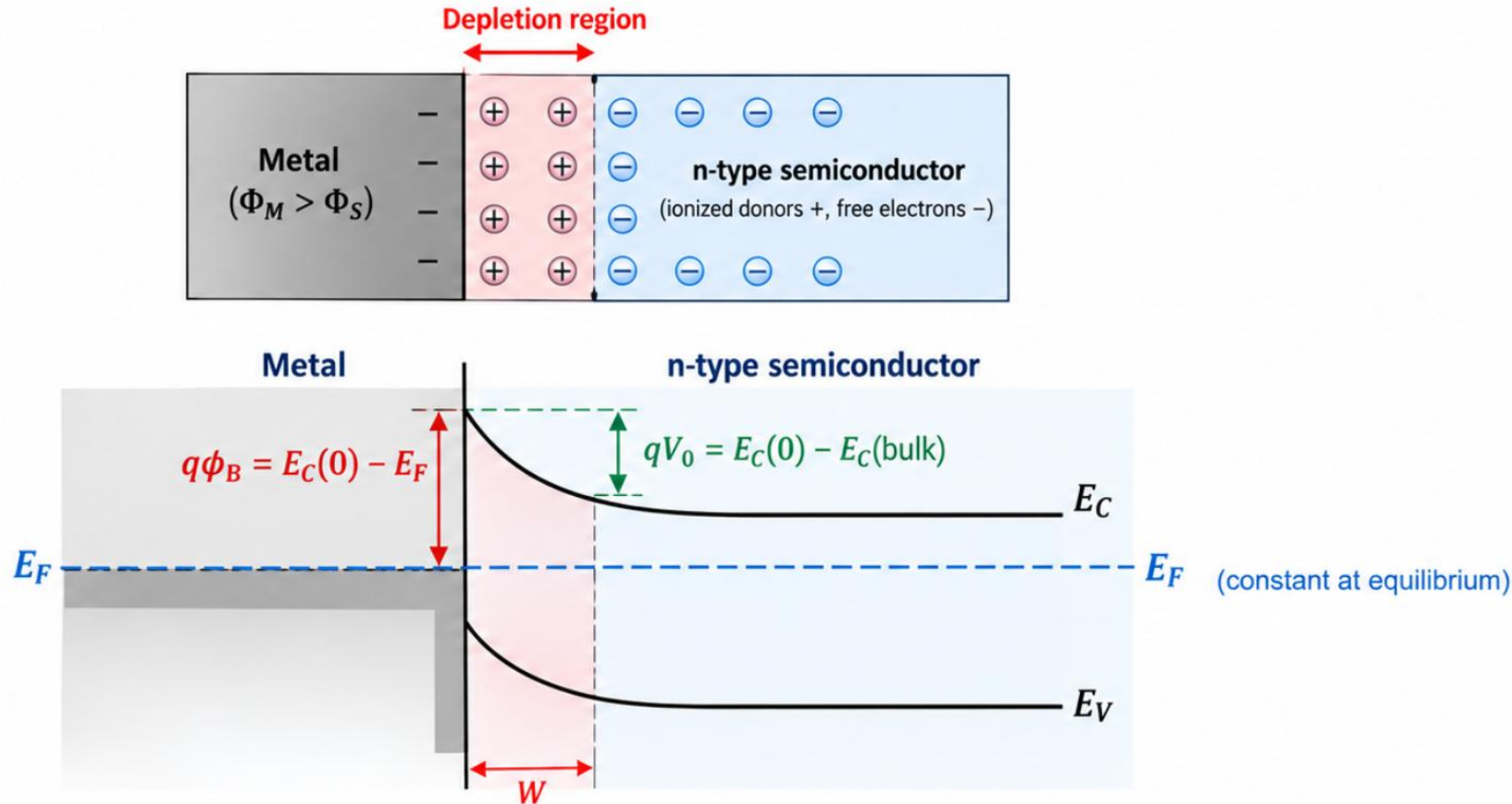
- How about this, as shown below?



- Since the body is buried inside the wafer, the body must be contacted from the top surface.
- Therefore, a metal contact is placed **ON** the body region to connect nMOS body → VSS and pMOS body → VDD. Is it done?
- However, simply placing a metal directly on the silicon body does not always provide a good electrical contact.** This is because a **metal–semiconductor interface** can introduce an energy barrier and a large contact resistance (See the next slide)
- Therefore, the body-contact region must be designed to behave as a low-resistance contact.

Why Can a Metal–Semiconductor Interface Be A Bad Contact?

Equilibrium Band Diagram

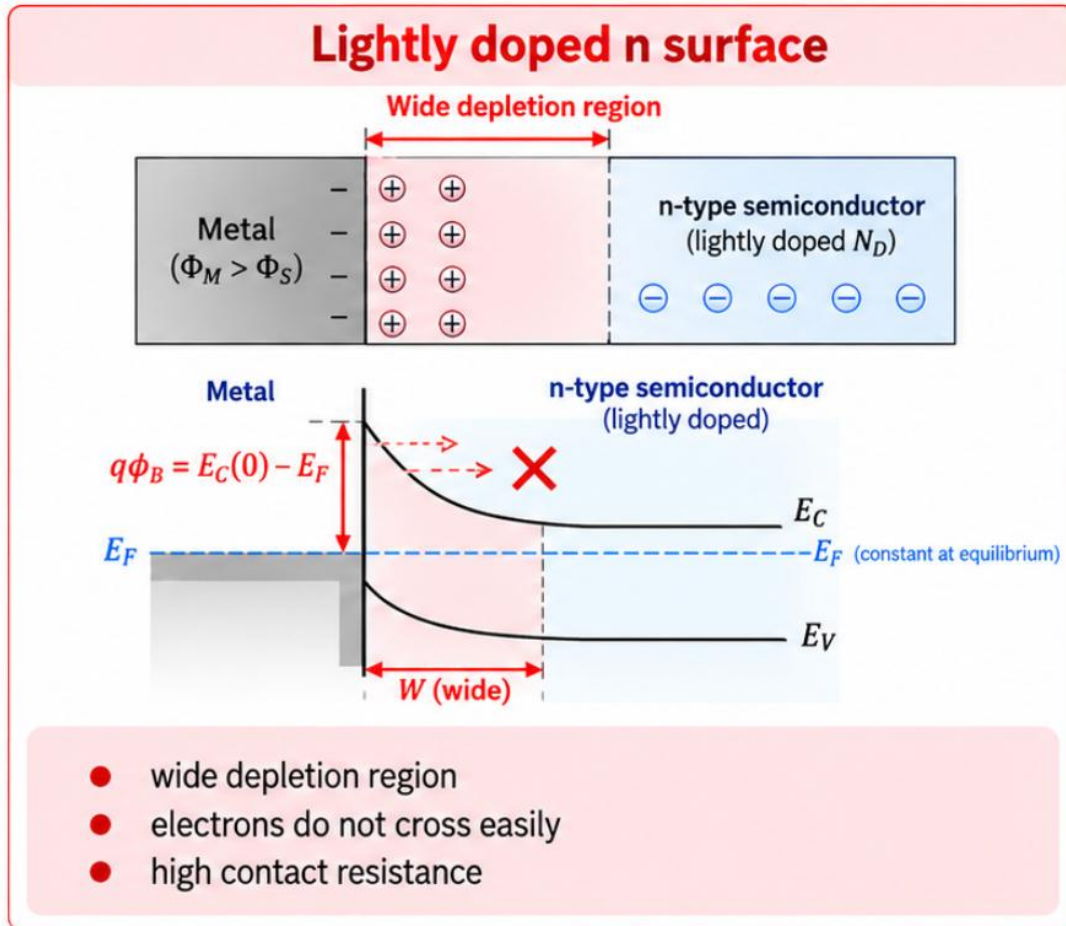


Schottky barrier → poor contact
large contact resistance, rectifying behavior

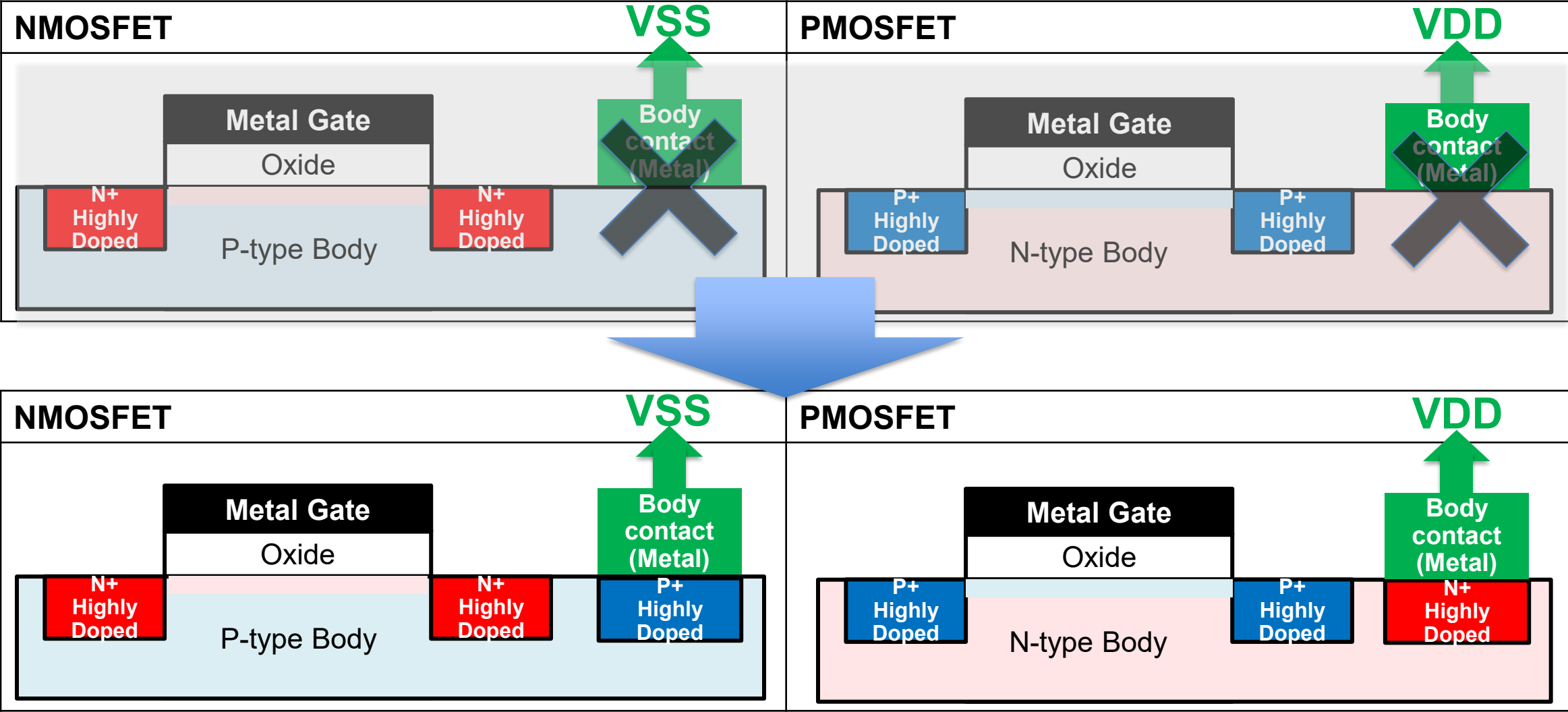
Rectifying behavior here means current flows only one direction as:
When External Voltage applied,

$V_{MS} \gg 0 \rightarrow$ Lowering Barrier $\rightarrow$ Current flows
 $V_{MS} < 0 \rightarrow$ Barrier is kept or even higher $\rightarrow$ Current not flows

How to Solve This Problem?



That is,



Background-3: Dynamic Random Access Memory

Data in Memory

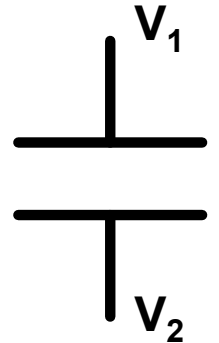
- Memory bitcell is a hardware unit that stores 1 bit data “0” or “1”
- Memory bitcell should ¹⁾hold/write/read ²⁾two possible states
 - 1) Hold: to retain the stored value over time
 - 2) Write: to change the memory bitcell into wanted state
 - 3) Read: to figure out what the memory bitcell state is
- Can you make memory bitcell?

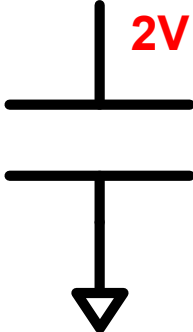
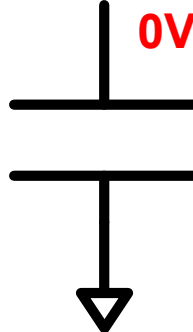
Holding Data with a Capacitor

- A capacitor stores electric charge on two conductive plates separated by an insulator.
- The stored charge determines the voltage difference between the plates:

$$Q = C(V_1 - V_2).$$

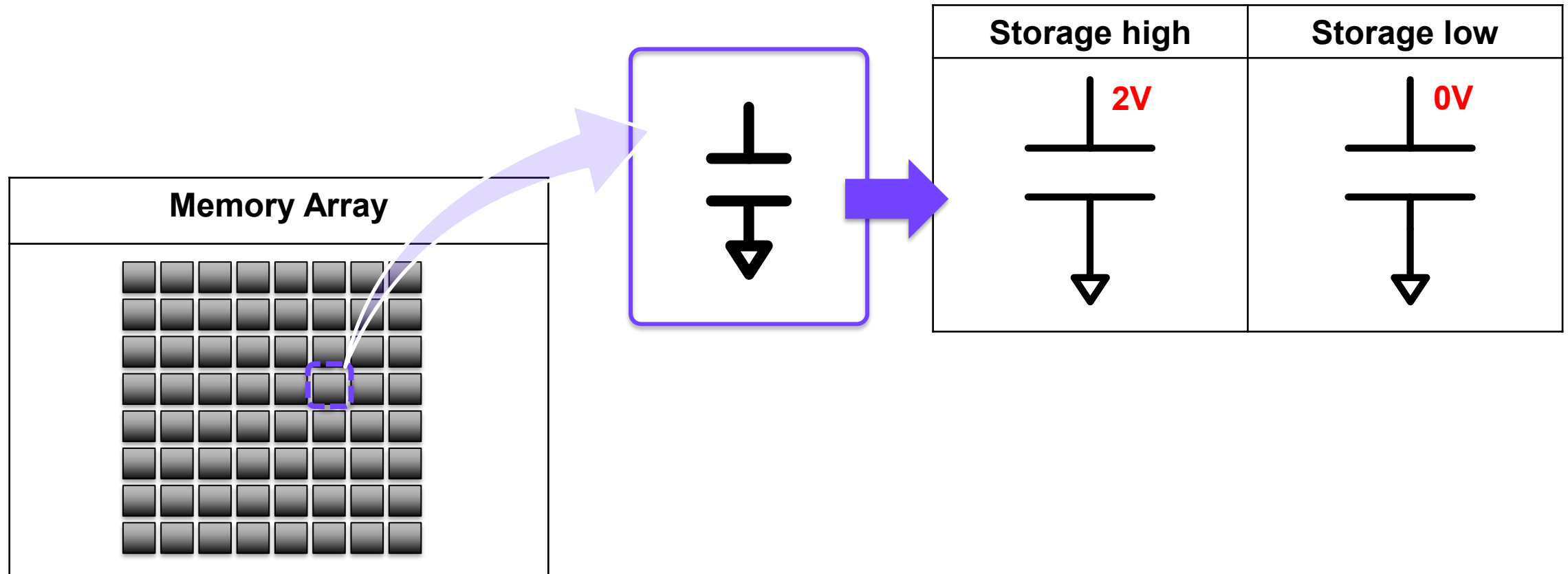
- If charge has no path to enter or leave, **the voltage difference is maintained.**
- With one terminal at a fixed voltage, the other terminal can hold a **high (“1”) or low (“0”) voltage**, storing one bit of data.
- We call this capacitor as a **memory bitcell**.



Storage high	Storage low
	

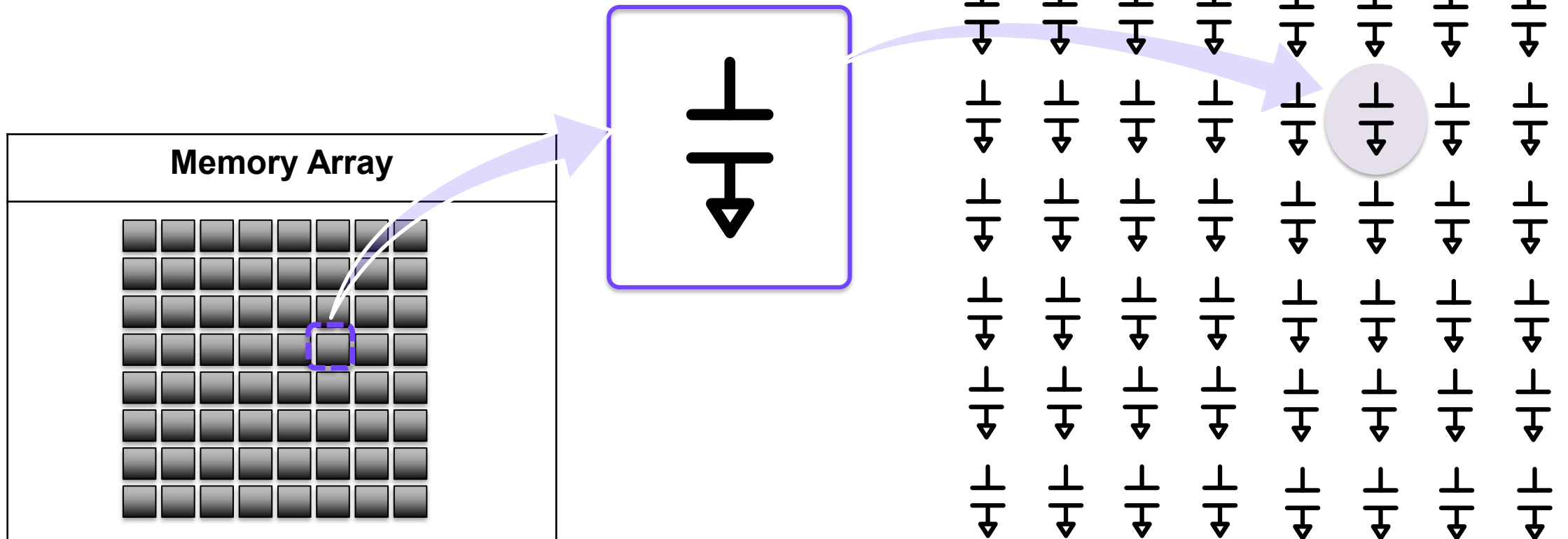
Storing Multiple Bits: Memory Array

- To store many bits within a limited chip area, bitcells are packed into a two-dimensional array.
- Each bitcell holds its own data, either “0” or “1”.
- **How can we select a desired bitcell to read or write its data?**



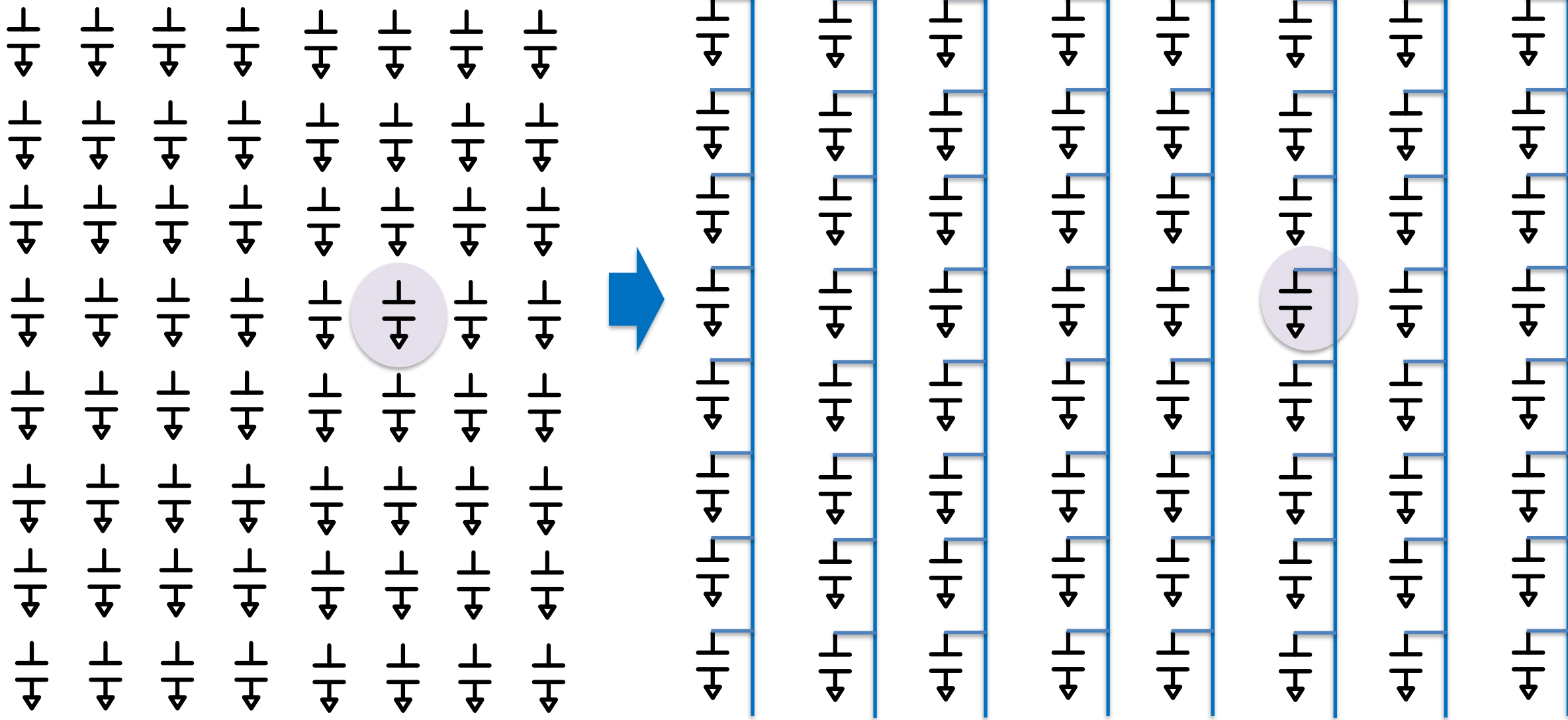
How to Access Specific Bitcell which is in Deep IN-side Array?

How can you access that specific bitcell, while it is located deep in the array??



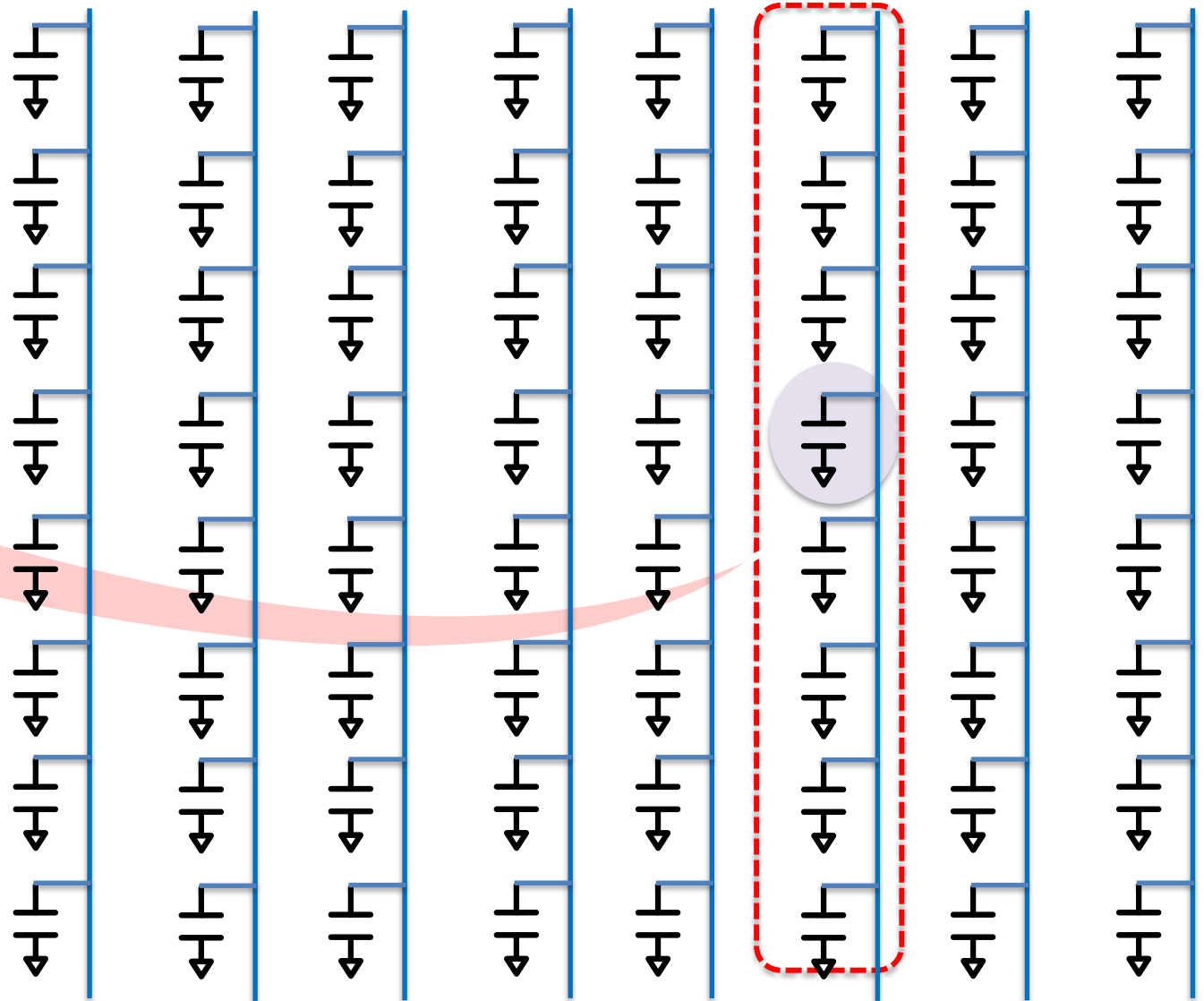
We Need A Line!

External observable lines are added as shown here!
We can observe the internal data outside



We Should Choose Only One!

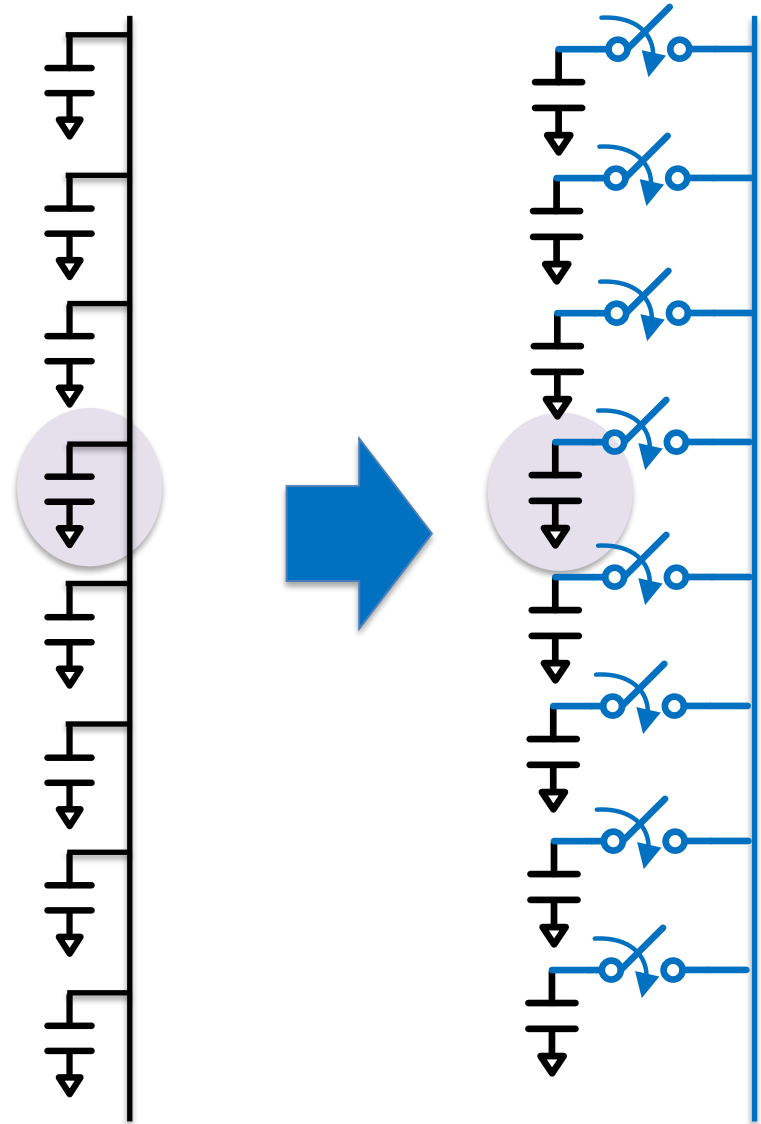
Among multiple capacitors
Only one capacitor (or bit-cell)
Should be selected to be accessed!



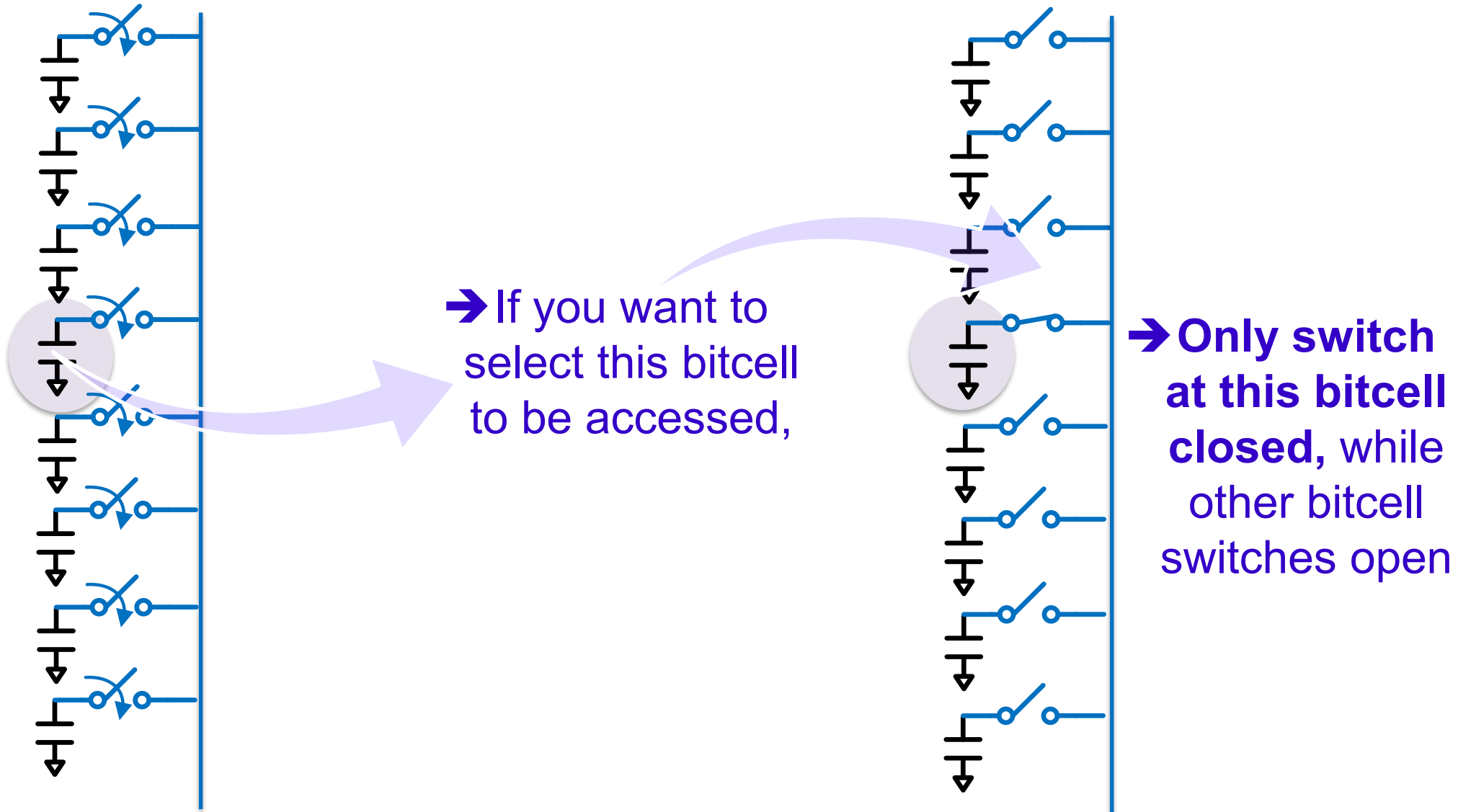
Only One Selection? We Need Switches!

Among multiple capacitors
Only one capacitor (or bit-cell)
Should be selected to be accessed!

→ We need to **switches** for
selectively connecting bitcells
according to our choice.

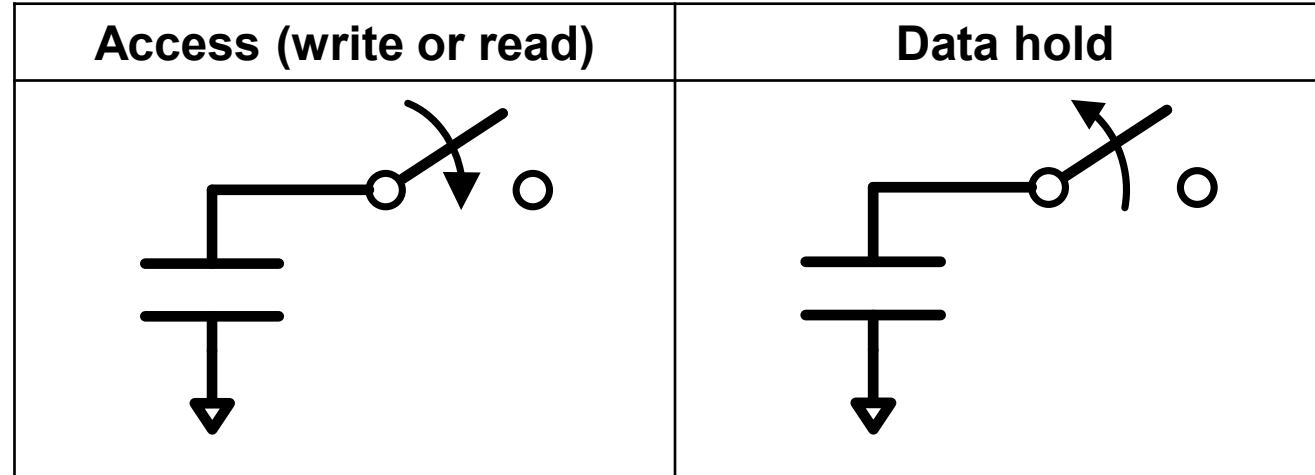


When You Access Only One Bitcell,



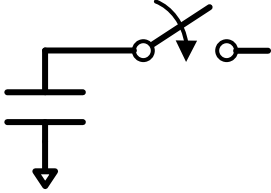
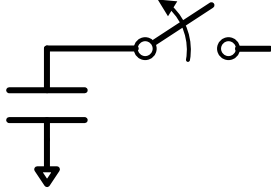
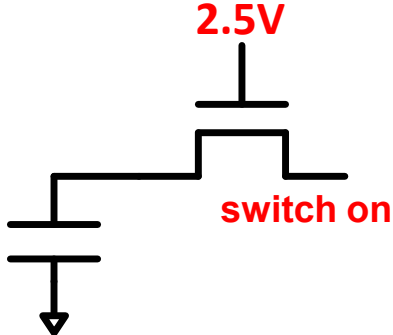
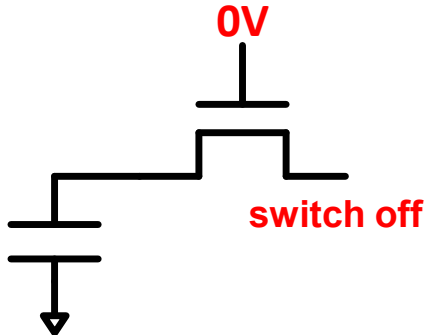
Accessing Stored Data with a Switch

- A switch controls the connection between the storage node and an external circuit.
- **ON — Read/Write:** Connect the capacitor to the external circuit to read or write data.
- **OFF — Hold:** Isolate the storage node to retain the stored charge.
- An ideal switch has **zero resistance when ON** and no **leakage current when OFF**.



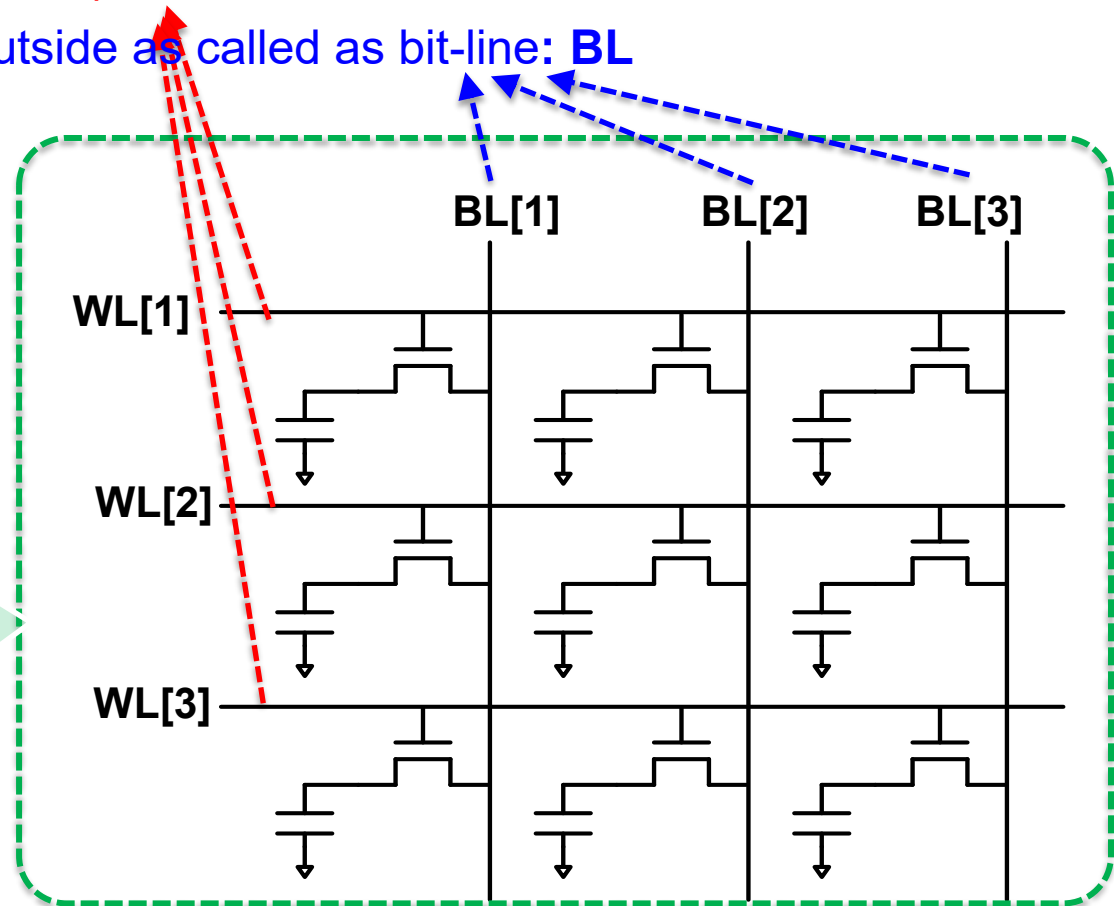
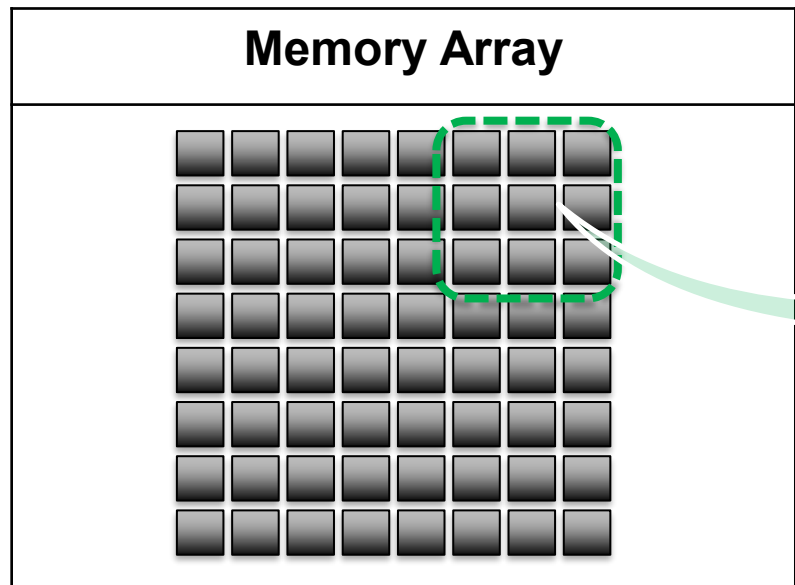
Implementing the Switch with an NMOS

- An NMOS transistor acts as a voltage-controlled switch.
- The gate voltage controls conduction between the source and drain.
- ON — Read/Write: Connect the storage node to the external circuit.
- OFF — Hold: Isolate the storage node to retain its charge.
- One NMOS transistor and one capacitor form a 1T1C DRAM bitcell.

Access (write or read)	Data hold
	
 2.5V switch on	 0V switch off

Controlling Multiple Bits in Memory Array

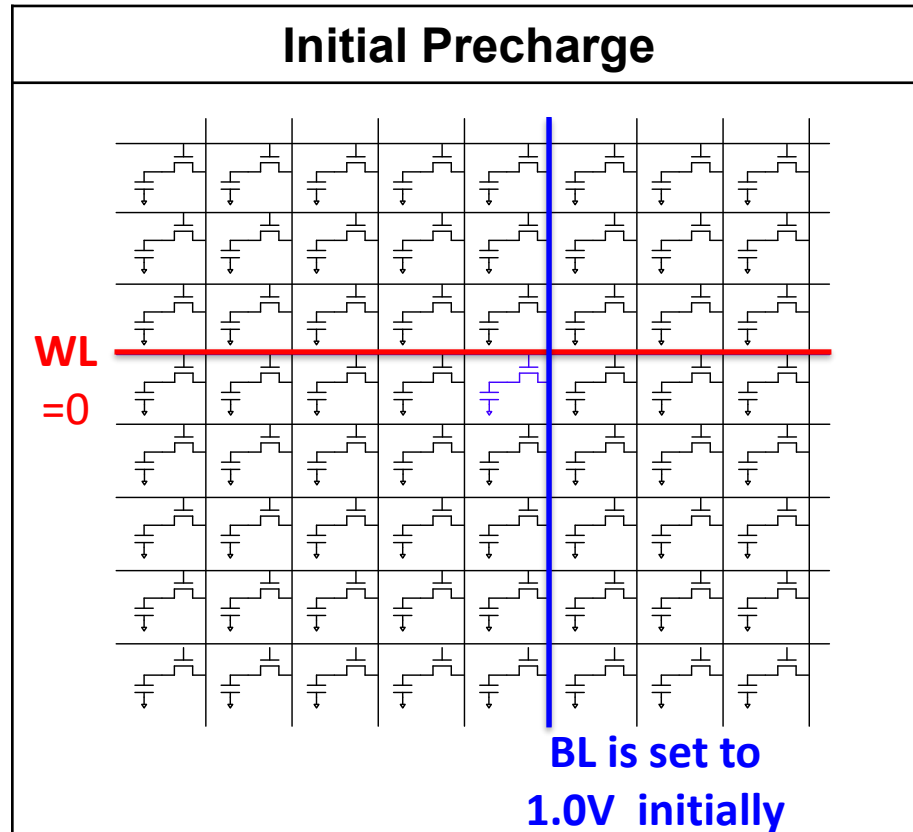
- To store many bits within a limited chip area, bitcells are packed into a two-dimensional array.
- Each bitcell holds its own data, either “0” or “1”.
- Horizontal connected signal which controls nMOS on/off is called as word-line: **WL**
- Vertical connected signal which connects data outside as called as bit-line: **BL**



Now, You Want to Find out the Data in the Selected Bitcell

: “Read” Operation of DRAM

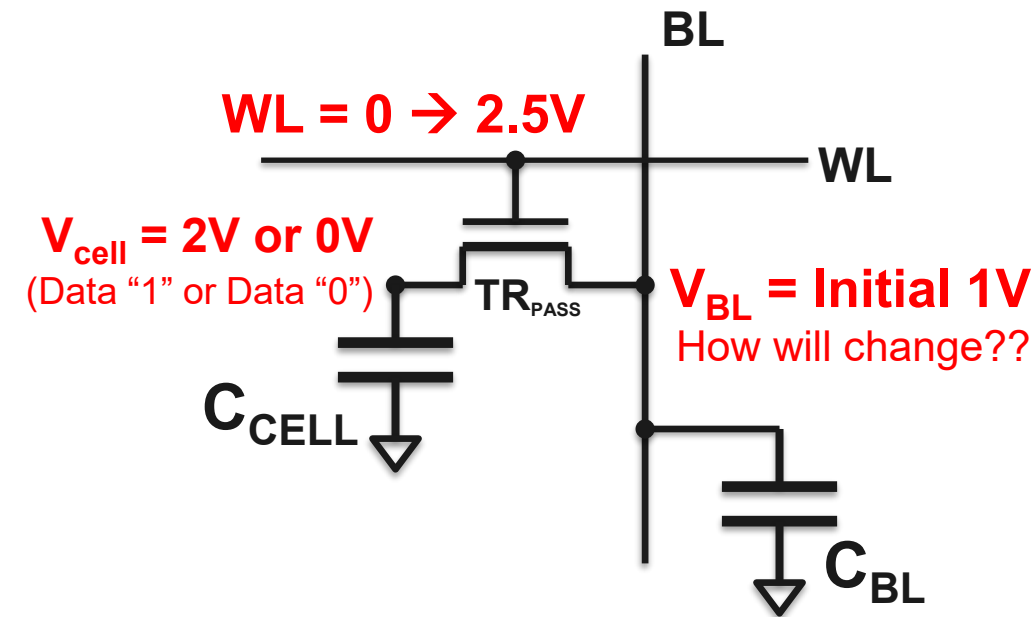
- WL turns on the access transistors in the selected row.
- Each cell in that row connects to its BL
- You will observe how BL changes from outside, and can determine the data between “1” or “0” according to how BL changes.



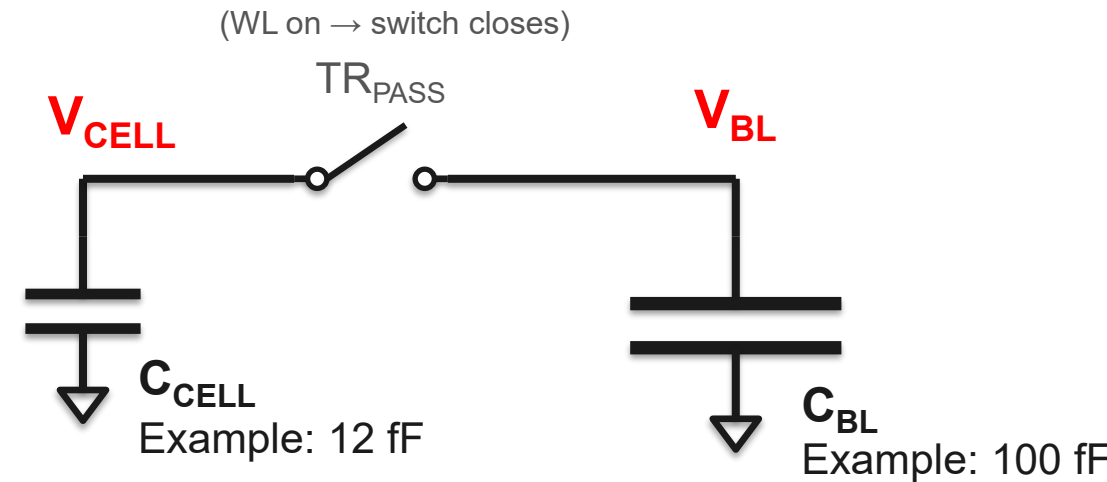
Reading a Bitcell

= Connecting Two Capacitors

- When WL turns on, the selected cell's capacitor is connected through the pass transistor to its BL.
- The BL is a long wire touching hundreds of cells, so it is **itself a capacitor**: for example $C_{BL} \approx 100$ fF against a cell's $C_{CELL} \approx 12$ fF, about eight to one in this example.
- A read is therefore: **a small charged capacitor meets a big one**. What voltage do they end up at?

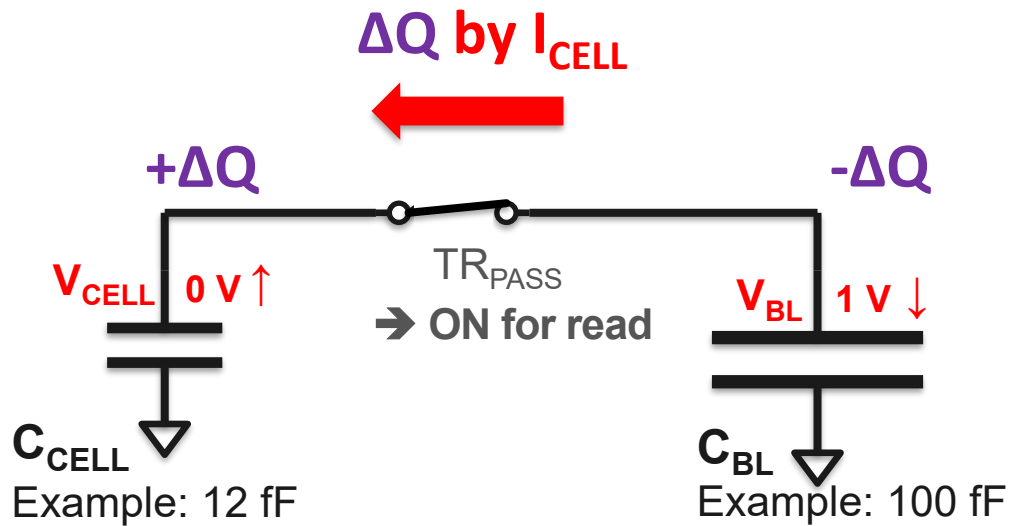


Seen from the storage node:
two capacitors
and a switch

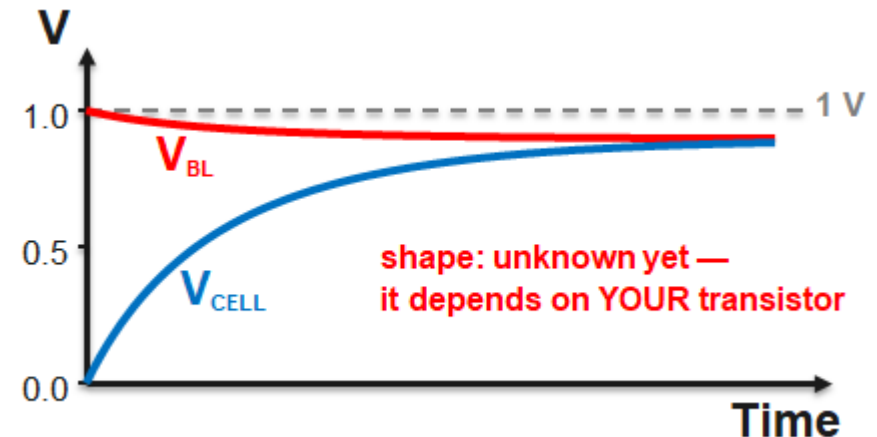
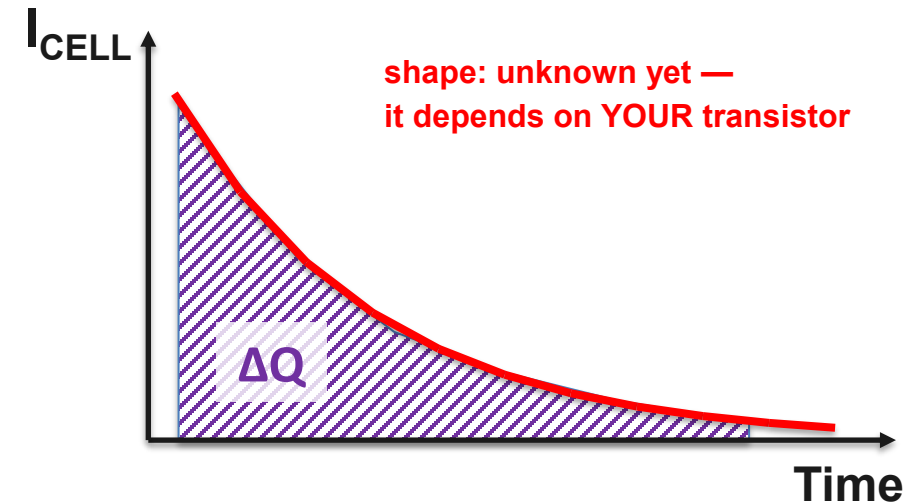


When the Switch Closes, Current Flows

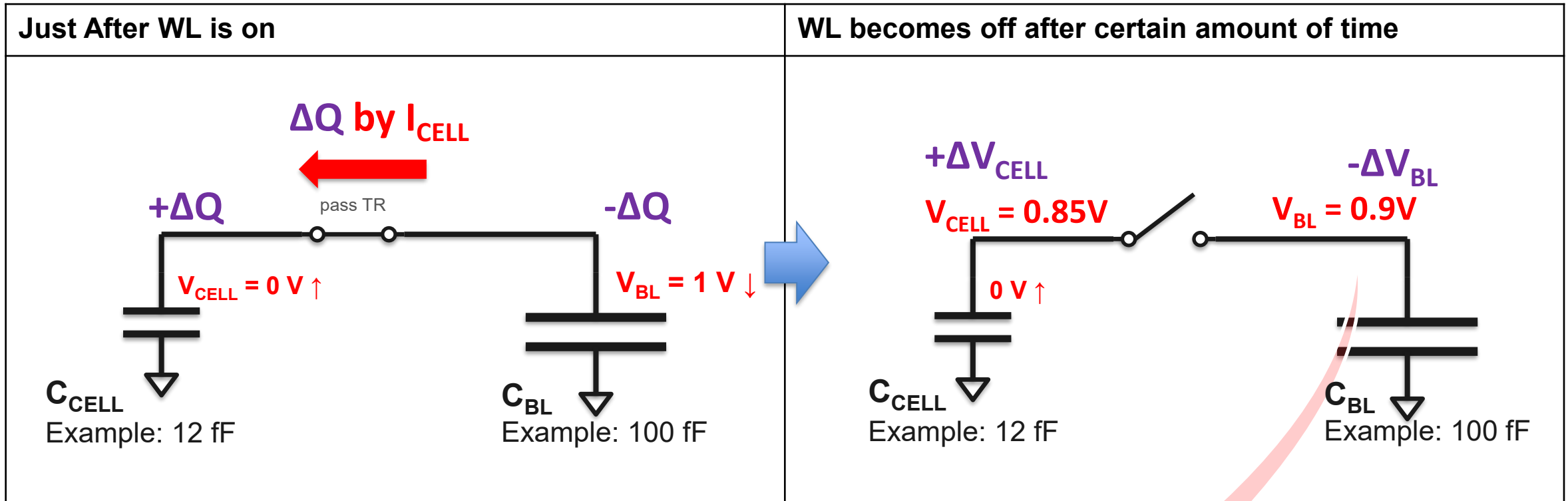
- The moment WL turns on, the pass transistor carries current (I_{CELL}) from the higher node to the lower
- When data “0” is stored, charges (Q) move from C_{BL} $\rightarrow$ C_{CELL}



Read “0”, just after WL turns on

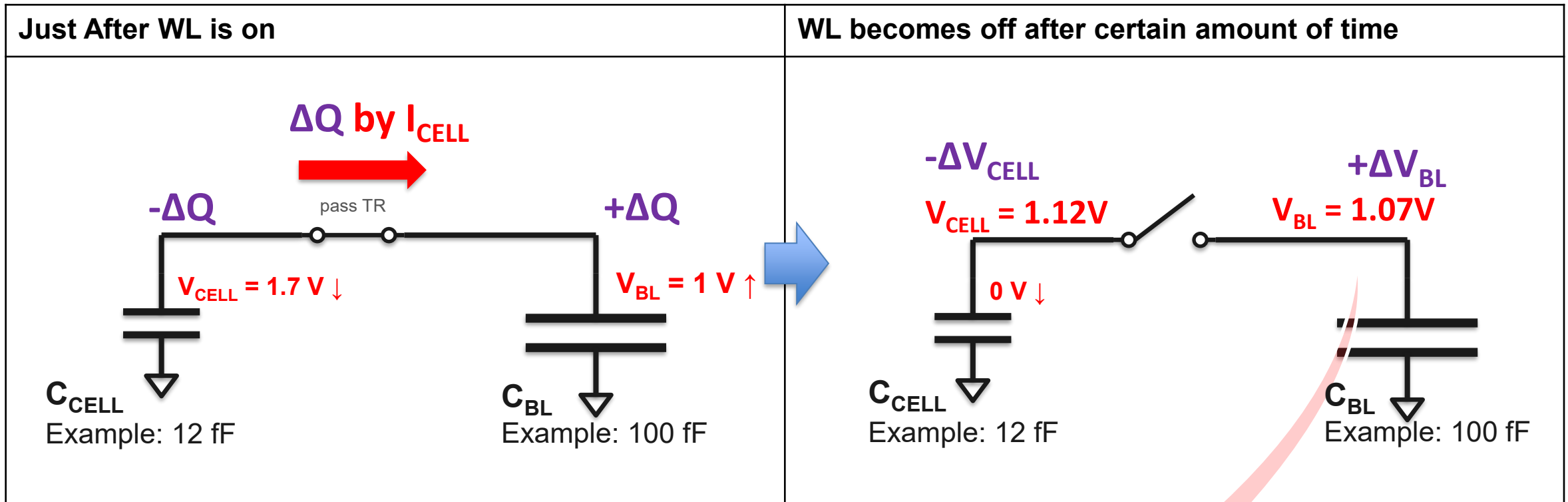


Resultant Read Procedure for Data “0”



At outside, you observe this data decrease at BL by 100mV (1V $\rightarrow$ 0.9V), you can say
“Oh, data 0 is stored inside the selected bitcell.” $\rightarrow$ This is read.

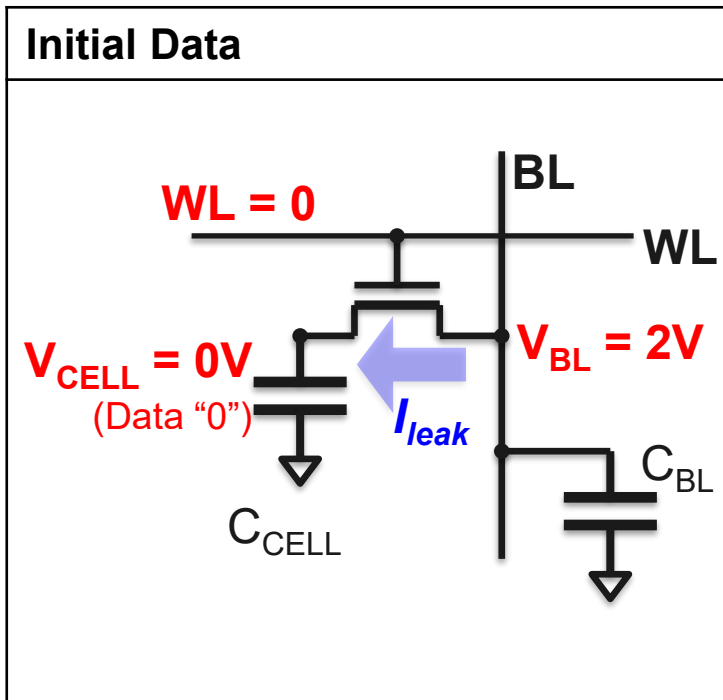
Resultant Read Procedure for Data “1”



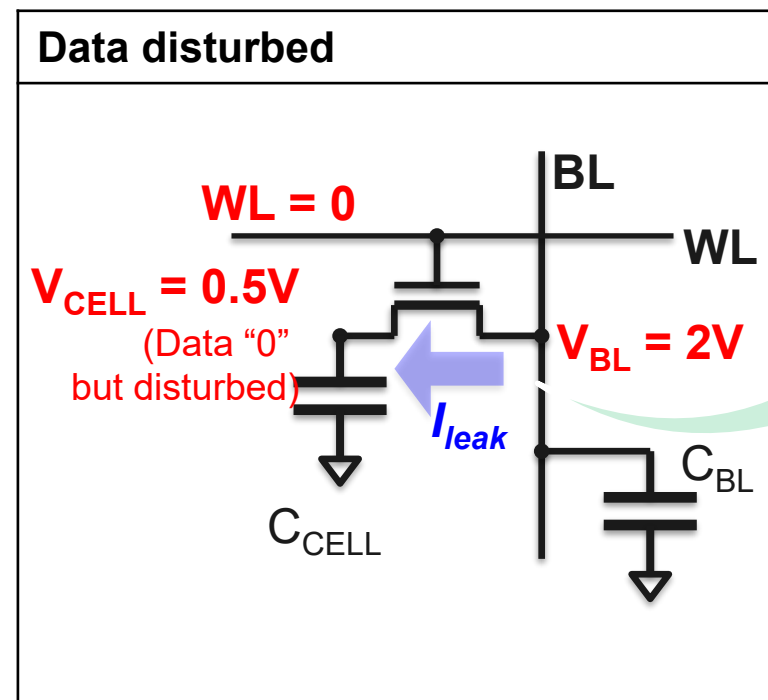
At outside, you observe this data increase at BL by 70mV (1V $\rightarrow$ 1.07V),
you can say
“Oh, data 1 is stored inside the selected bitcell.” $\rightarrow$ This is read.

Data Retention Issue

- When $WL = 0$, data is not accessed, so CCELL should keep the data unchanged.
- However, the leakage current (off-current of transistor) makes the VCELL change inevitably.
- Note that V_{BL} can swing from 0~2V due to write condition. For data "0" retention shown below, $V_{BL} = 2V$ case should be considered as the worst cases.



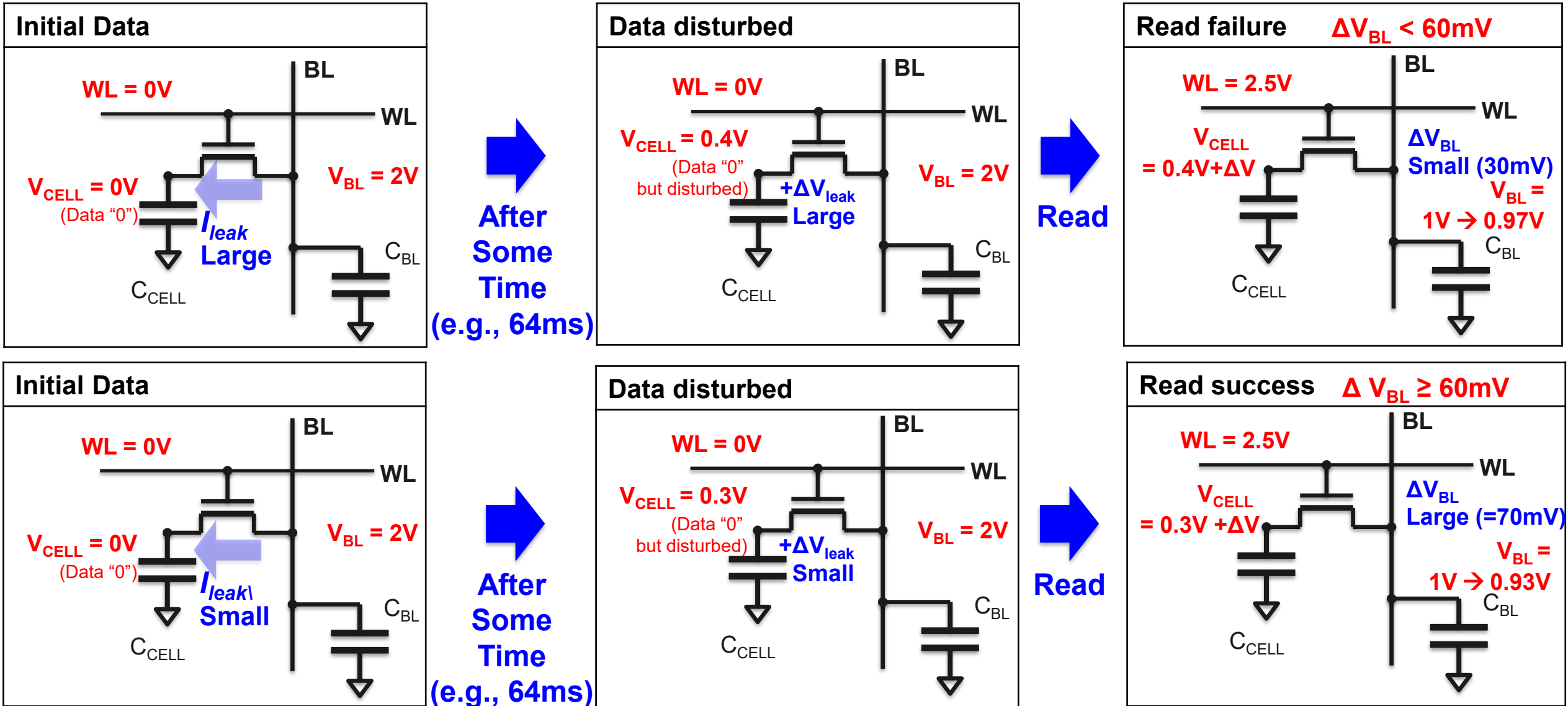
➡
**After
Some
Time**
(e.g., 64ms)



If you perform read at this condition ΔV_{BL} will be decreased not by 100mV, but much less amount (like only 30mV or 10mV...)

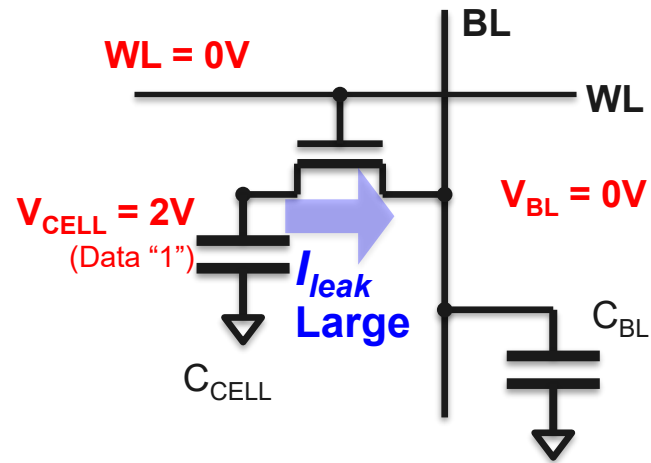
➔ **Undesirable!** because we need some minimum ΔV_{BL} (like 60mV) for stable read operation (called **read margin**)

If Required Read Margin $\geq 60\text{mV}$ for Data "0" With Retention Time of 64mS



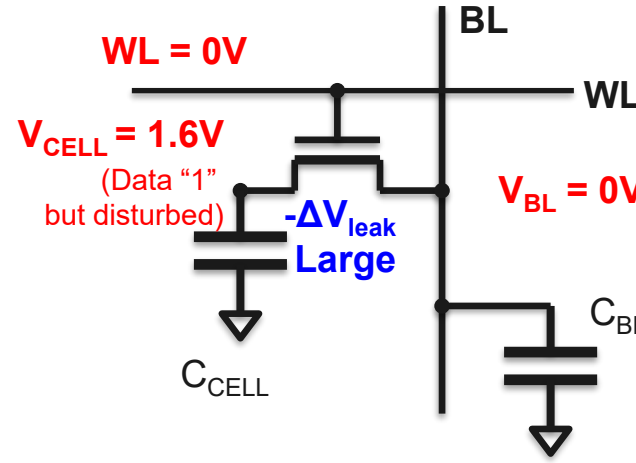
If Required Read Margin $\geq 60\text{mV}$ for Data "1" With Retention Time of 64mS

Initial Data



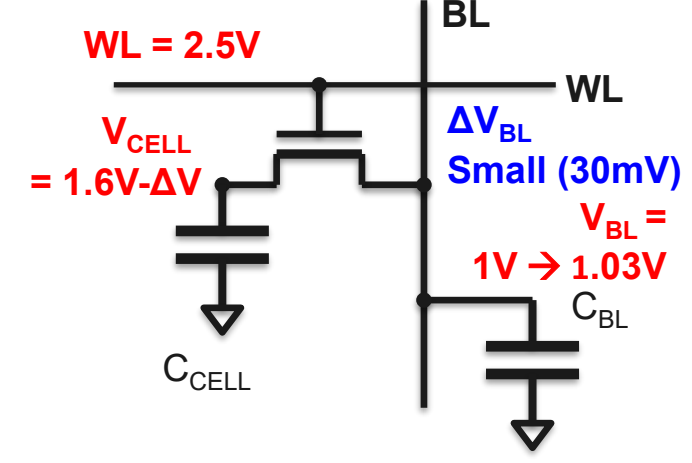
After
Some
Time
(e.g., 64ms)

Data disturbed

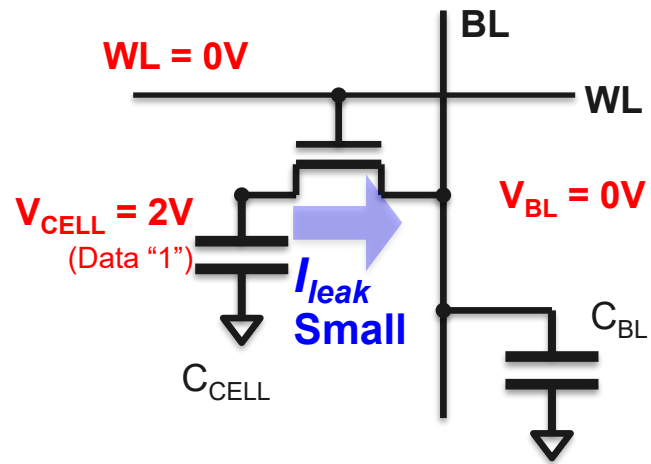


Read

Read failure $\Delta V_{\text{BL}} < 60\text{mV}$

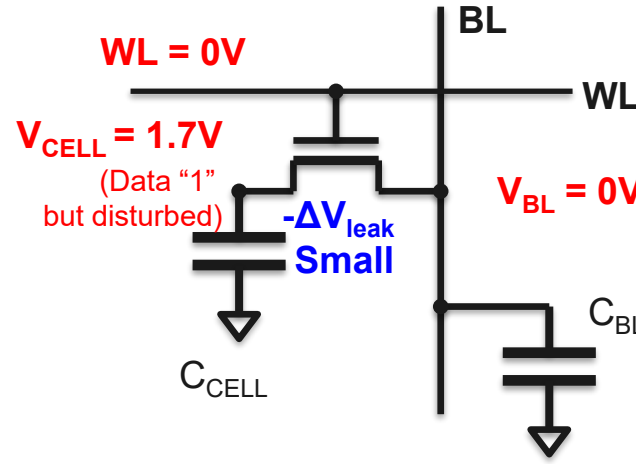


Initial Data



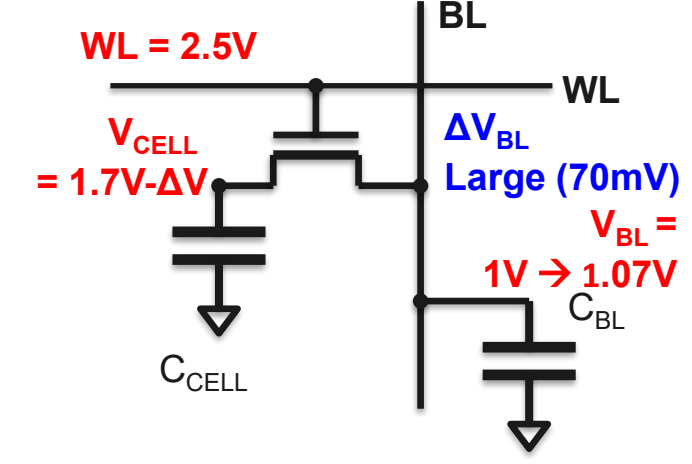
After
Some
Time
(e.g., 64ms)

Data disturbed



Read

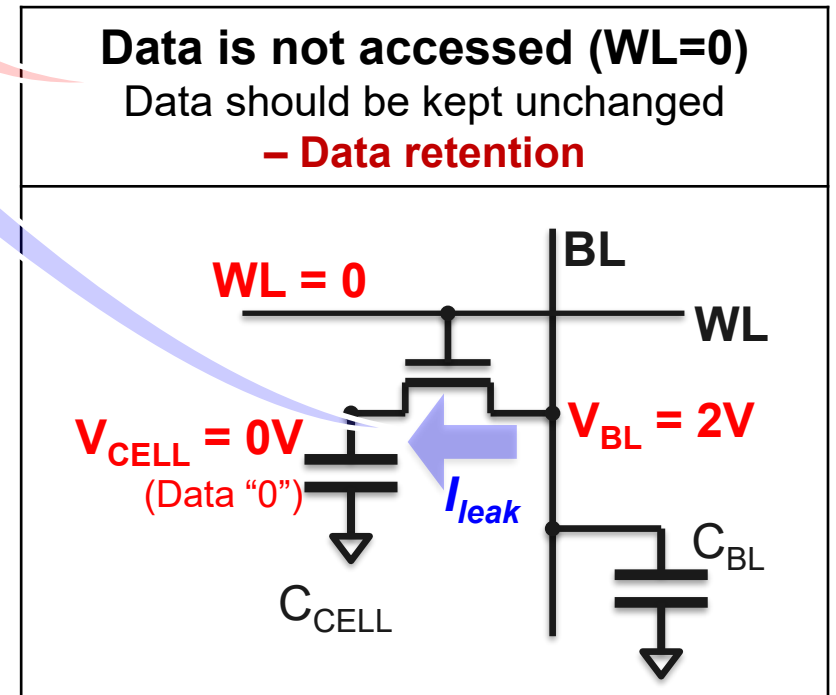
Read success $\Delta V_{\text{BL}} \geq 60\text{mV}$



Why Does DRAM Need Sufficient Capacitance?

1. **A larger cell capacitance improves better read margin.**
→ For the same initial cell voltage and bitline capacitance, a larger cell capacitance produces a larger BL voltage change.
2. **When WL = 0, and data is not accessed, a larger cell capacitance improves better data retention.**
→ For the same leakage current, a larger capacitance slows the change in storage voltage:

$$\left| \frac{dV}{dt} \right| = \frac{|I_{leak}|}{C_{cell}}$$



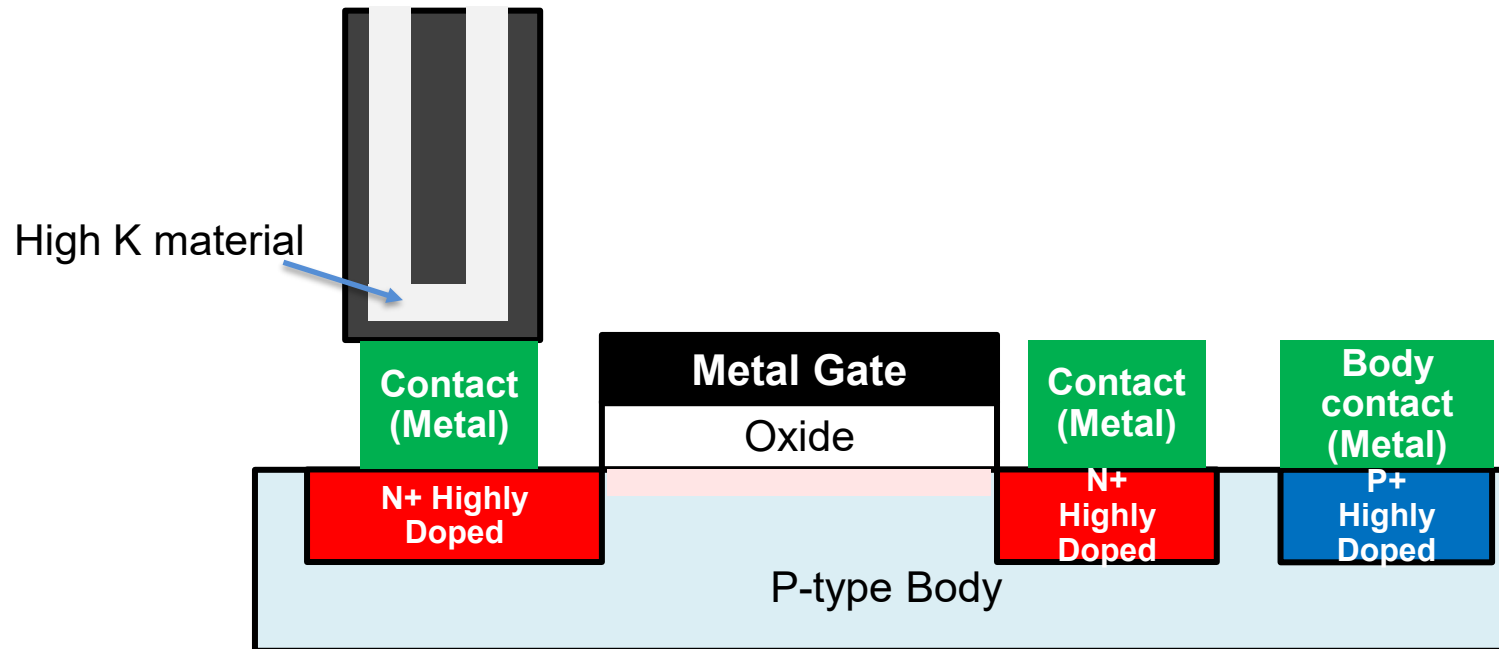
Vertical Capacitors for Higher Memory Density

- A vertical capacitor provides a large capacitance within a small footprint.
- Capacitance increases with electrode surface area:

$$C \approx \frac{\epsilon A}{t},$$

(ϵ : permittivity, A : area, t : thickness)

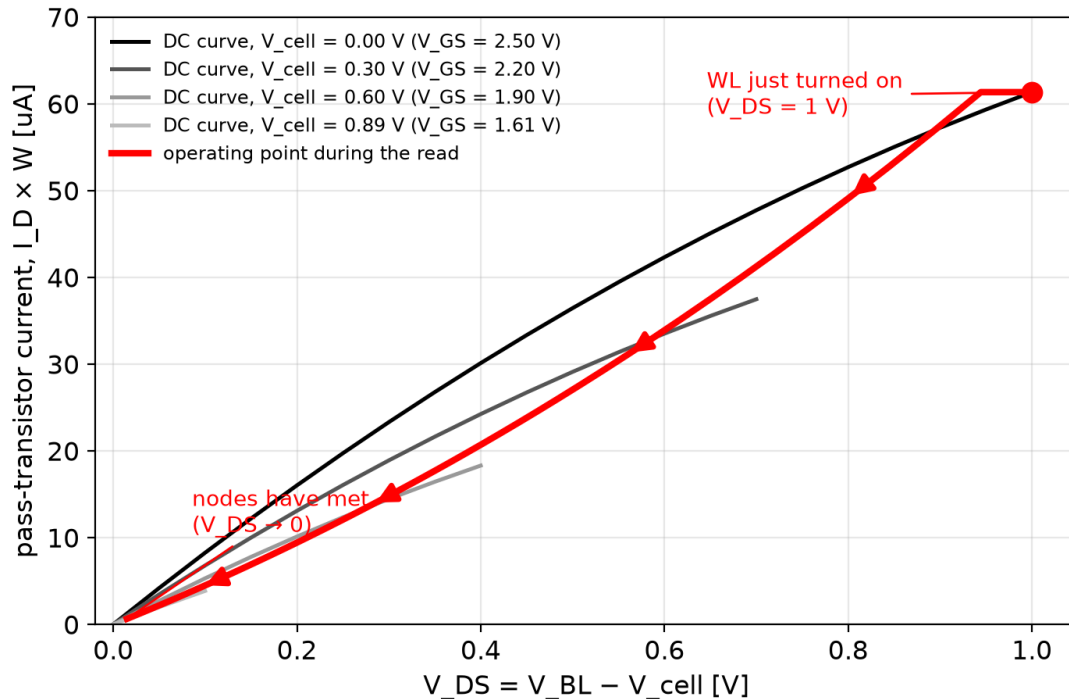
- Extending the electrodes vertically increases their facing surface area without substantially increasing the footprint.
- A thin dielectric separates the storage electrode from the surrounding plate electrode.



Background-4: Time-based Simulation

At Every Instant, the Transistor Obeys Its DC I–V Curve

- Your DEVSIM simulator is a **DC solver**: fix the voltages, get one current. It knows nothing about time.
- That is all we need, because at any instant $I(t) = I_D(V_{GS}(t), V_{DS}(t))$ — the current is simply the DC value at the voltages of that instant.
- As the read proceeds V_{DS} shrinks (and V_{GS} too, since the source node rises): the **operating point slides down the I_D – V_{DS} curves toward the origin**.

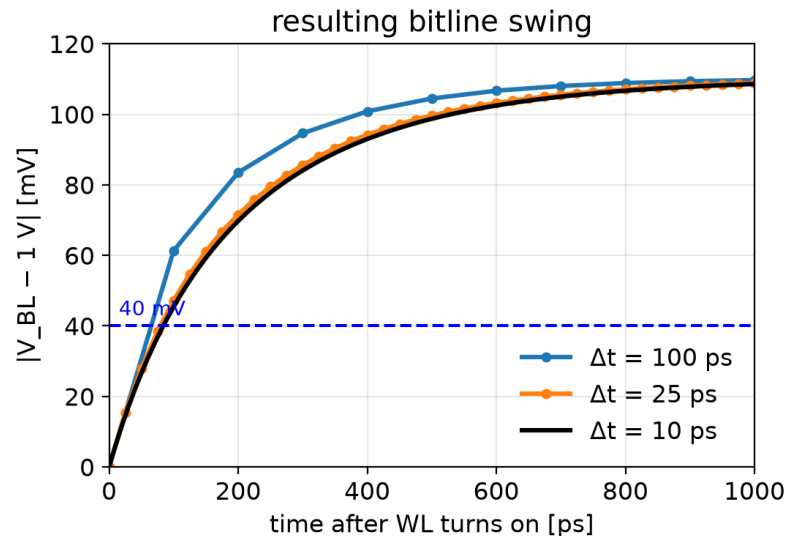
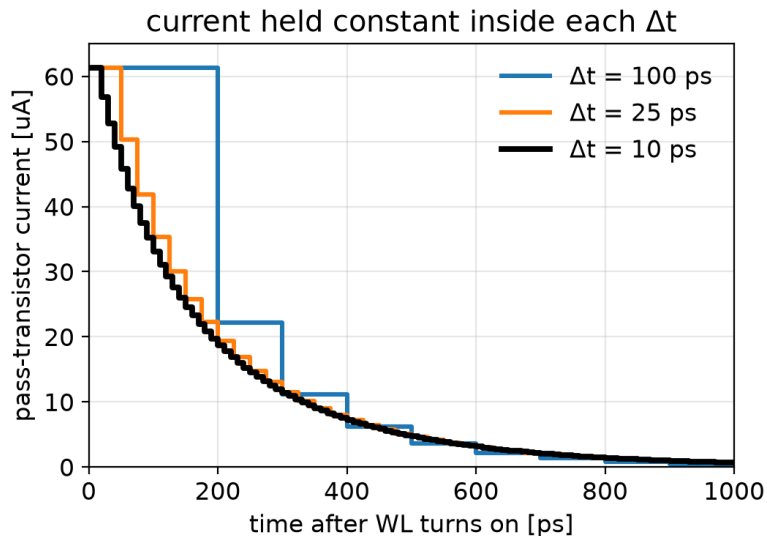


- **Gray curves:** ordinary DC I_D – V_{DS} sweeps of the pass transistor, one per storage-node voltage (your simulator can produce these).
- **Red path:** where the transistor actually operates during a “0” read, from $V_{DS} = 1$ V down to 0.
- If we knew the voltages at each moment, we could read the current off these curves. **The missing piece:** how the voltages move with time. That comes from the capacitors — next slide.

Follow Time in Small Steps: Δt

- The capacitors link current and voltage: $I = C \cdot dV/dt$. Over a short step Δt the current hardly changes, so $\Delta Q = I \cdot \Delta t$, $\Delta V_{\text{cell}} = +\Delta Q / C_{\text{STORE}}$, $\Delta V_{\text{BL}} = -\Delta Q / C_{\text{BL}}$.
- **Loop:** DC solve at the present voltages $\rightarrow I \rightarrow$ move $\Delta Q \rightarrow$ new voltages $\rightarrow$ DC solve again ... until the decision time (1 ns).
- **How small must Δt be?** Small compared with the time the voltages take to change (cell RC ≈ 0.2 ns here). Shrink Δt until the result stops changing: **$\Delta t = 10$ ps** (100 steps to 1 ns) is enough.

Read "0": same device, three step sizes



- Inside each Δt the current is held constant $\rightarrow$ the staircase.
- $\Delta t = 100$ ps: the first step moves too much charge $\rightarrow$ visibly off.
- **$\Delta t = 25$ ps and 10 ps almost coincide $\rightarrow$ converged. Use 10 ps.**
- This is what SPICE does with .tran — here the “device model” is your own DEVSIM device.

Project Part 1

Design an nMOS device in DEVSIM that satisfies the given electrical specifications.

Start from the simulator you built in HW#1, change the structure, and prove every spec with your own simulation.

Electrical Specifications ($V_{DD} = 2\text{ V}$, W-normalized units per μm)

Item	Condition	Criterion
V_{th}	constant-current definition: $I_D = 1\text{e-}7\text{ A}/\mu\text{m}$ at $V_D = 0.05\text{ V}$	$0.45 \pm 0.05\text{ V}$
I_{on}	$V_G = V_D = 2\text{ V}$, 300 K	$\geq 450\text{ }\mu\text{A}/\mu\text{m}$
I_{off}	$V_G = 0$, $V_D = 2\text{ V}$, 300 K	$\leq 1\text{ pA}/\mu\text{m}$
SS	$V_D = 0.05\text{ V}$, average over $I_D = 1\text{e-}10 \dots 1\text{e-}8\text{ A}/\mu\text{m}$	$\leq 75\text{ mV/dec}$
High-temperature I_{off}	$V_G = 0$, $V_D = 2\text{ V}$, 125 °C (398 K)	$\leq 100\text{ pA}/\mu\text{m}$
DIBL	$(V_{th}\text{ at } V_D = 0.05\text{ V} - V_{th}\text{ at } V_D = 2\text{ V}) / 1.95\text{ V}$	$\leq 30\text{ mV/V}$
Body effect	V_{th} increase when $V_B = -0.5\text{ V}$ ($V_D = 0.05\text{ V}$)	$\leq 0.08\text{ V}$
Oxide field	$E_{ox} = V_{DD} / t_{ox}$	$\leq 5\text{ MV/cm}$

What This Means for You

- All eight items are measured on the **same device** — one structure, one doping profile, one gate material.
- Some items **cannot be measured with the HW#1 code as distributed**. Find out which physics is missing, add it to your simulator, and verify the implementation before you trust the number.
- The conditions are deliberately in tension: what helps I_{on} tends to hurt I_{off} , E_{ox} or DIBL. There is a solution; the job is to find the balance.
- Report every number with the exact bias, temperature and definition above, in $\text{A}/\mu\text{m}$ (2-D result per μm of width).

Important Notes

- Deliverable for Part 1: your device file part1_[studentID].devsim (structure, doping, materials) + report. The TA re-runs the file with the TA physics; the re-run decides pass/fail, so keep some margin.
- Fixed for Part 1: $L_g \geq 0.3\text{ }\mu\text{m}$, gate dielectric = SiO_2 , $T = 300\text{ K}$ unless the item says otherwise, no changes to code constants (mobility, ramps) — replacing a constant with a physical model is allowed.

You can change the device dimensions, doping profile, and gate material.

Use these design knobs to meet all electrical specifications.

Design Parameters and Allowed Ranges

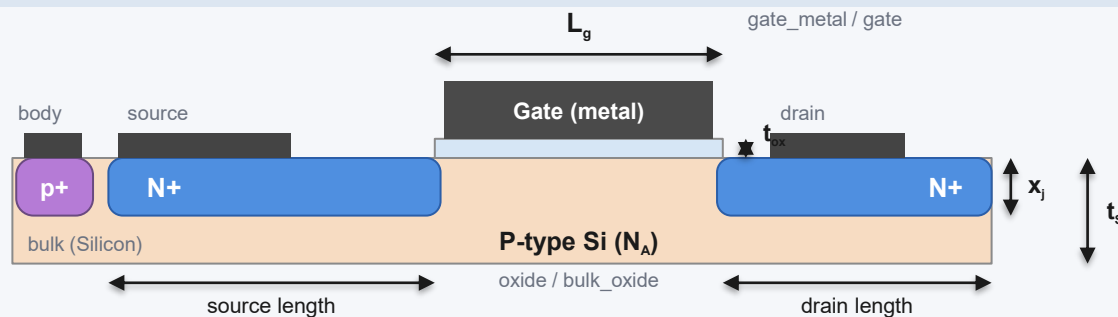
Parameter	Symbol	What you can change / how it acts	Allowed range
Gate length	L_g	longer or shorter (V_{th} , I_{on} , I_{off} , DIBL)	$\geq 0.3 \mu\text{m}$
Gate oxide thickness	t_{ox}	thicker or thinner (V_{th} , I_{on} , E_{ox})	$\geq 4 \text{ nm}$
Junction depth	x_j	deeper or shallower S/D junctions (DIBL, leakage)	$0.02 - 0.25 \mu\text{m}$
Silicon thickness	t_{Si}	substrate depth in the simulation domain	$0.3 - 2 \mu\text{m}$
Source/drain length	L_{SD}	lateral S/D extension (series resistance)	$0.2 - 1 \mu\text{m}$ each
Body doping (p)	N_A	concentration and spatial profile	$10^{14} - 10^{17} \text{ cm}^{-3}$
S/D doping (n+)	N_D	concentration and spatial profile	$10^{18} - 10^{21} \text{ cm}^{-3}$

Gate Material Options (effective work function on SiO_2)

Gate	Φ_m [eV]	Gate	Φ_m [eV]	Gate	Φ_m [eV]
n+ poly-Si	4.05	TaN	4.45	Ni	5.10
Al	4.10	W	4.60	p+ poly-Si	5.15
Ta	4.25	TiN	4.65	Pt	5.30
Ti	4.33	Mo	4.70		

Higher work function $\rightarrow$ higher V_{th} ; lower $\rightarrow$ lower V_{th} . Write the material name in the gate_metal region of your file; the TA applies the table value. Check what the HW#1 code assumes for the gate.

Illustration of Design Knobs (2-D cross-section, names used in the file)



Hints for Design (general device physics only)

To increase V_{th}

- higher work-function metal
- thicker t_{ox} , higher N_A

To increase I_{on}

- shorter L_g ($\geq 0.3 \mu\text{m}$), thinner t_{ox} ($\geq 4 \text{ nm}$)
- higher N_D , shorter S/D (less series resistance)

To reduce I_{off}

- higher V_{th} (metal, t_{ox} , N_A)
- watch what happens at 125°C

To reduce DIBL

- longer L_g , shallower x_j , higher N_A near the channel

Important Notes

- You may modify the structure, mesh and doping profile — not just a few numbers. Keep the required region/contact/interface names (see the submission pages).
- Keep the gate dielectric as SiO_2 and use the work-function table above. Do not arbitrarily change mobility or numerical settings; replacing a constant with a physical model is allowed.

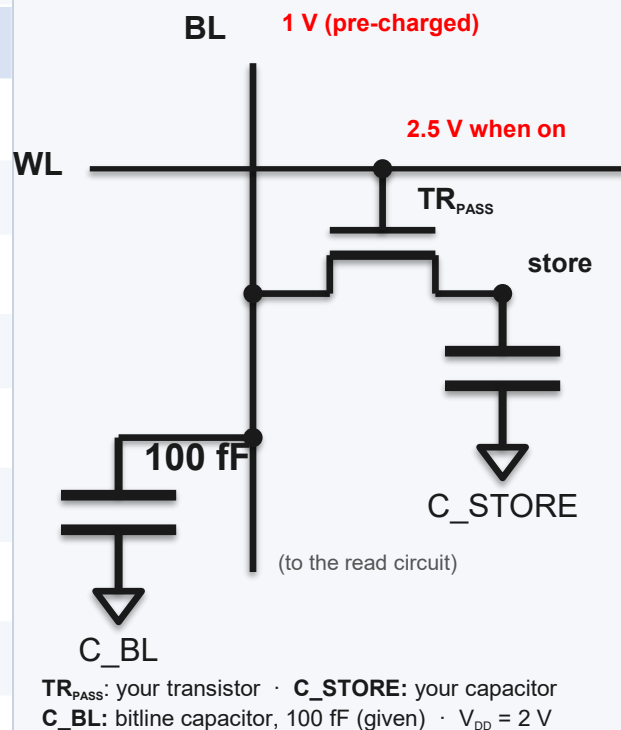
Design a 1T1C DRAM bitcell that reads within 0.5 ns and retains data ≥ 64 ms — the longer, the better.

Build both devices as one DEVSIM structure. Both the read and the retention are judged by Δt -stepped time-domain (transient) simulations.

Operating Conditions (given)

Quantity	Value	Note
V_{DD} / BL pre-charge	2 V / 1 V	BL is set to $V_{DD}/2$ before every read (pre-charge)
WL high (V_{PP})	2.5 V	gate of the pass TR during read
C_{BL}	100 fF	ideal bitline capacitor
C_{STORE}	your design, ≤ 20 fF	extracted from your capacitor structure (plate dQ/dV)
Minimum BL change to read, V_{offset}	60 mV	must be reached 0.5 ns after WL rises (decision time)
Initial stored levels	"1" = 2 V, "0" = 0 V	no write simulation; read and retention start from these levels
Temperature / body bias	125 °C / $V_b = -0.5$ V	all Part 2 simulations
Pass TR geometry	$L_g = 0.3$ μ m fixed, $W = 100$ nm	cell area; other knobs free within Part 1 ranges
Oxide field	$E_{ox} = 2.5$ V / $t_{ox} \leq 5$ MV/cm	WL = 2.5 V sits on the gate

The Cell You Design (1T1C)



Pass / Fail Criteria

Read "0" and read "1"

$|V_{BL} - 1 \text{ V}| \geq 60 \text{ mV}$ at $t = 0.5 \text{ ns}$ after WL turns on (decision time), for both stored states. Computed with the Δt -stepped transient using your DC device (see background).

Retention

WL = 0 V, 125 °C, $V_b = -0.5$ V. Hold data "0" with $V_{BL} = 2$ V and data "1" with $V_{BL} = 0$ V. Follow the cell voltage with a Δt -stepped transient; after 64 ms the same read must still give ≥ 60 mV. Retention time = when the margin drops below 60 mV; longer is better.

Reliability

$E_{ox} \leq 5$ MV/cm with 2.5 V on the gate $\rightarrow t_{ox} \geq 5$ nm. Capacitor dielectric field $1 \text{ V} / t_{diel} \leq 4$ MV/cm.

Why it is hard: low V_{th} reads fast but leaks; high V_{th} holds data but moves less charge in 0.5 ns, so the "1" read margin shrinks. The Part 1 reference fails retention at 125 °C. Tune gate metal, doping and capacitor together.

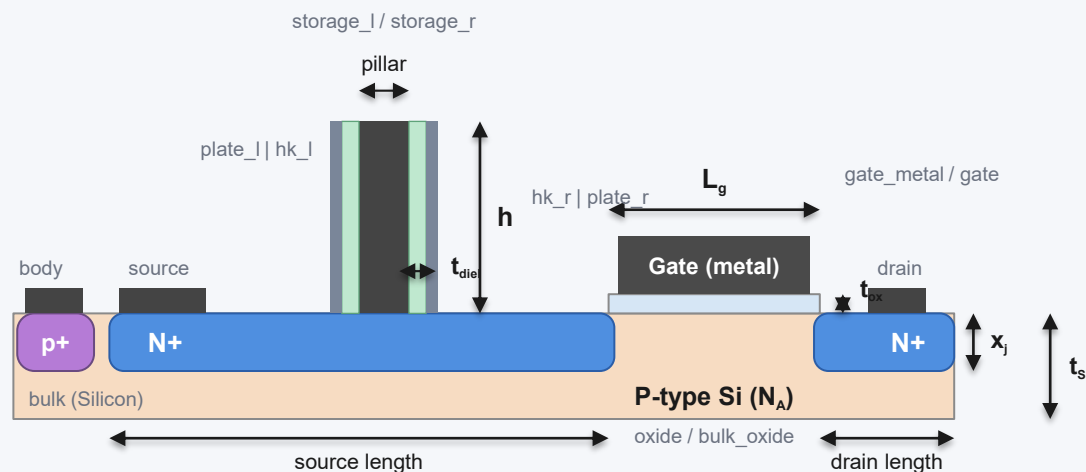
Important Notes

- Deliverable for Part 2: part2_[studentID].devsim containing the pass transistor and the capacitor in one structure + report with the transient read curves, the retention estimate and your C_{STORE} extraction.
- The TA loads your file, attaches the TA physics and repeats the same read/retention procedure. Results within the TA's tolerance count; leave margin.

You design the pass transistor and the storage capacitor as one 2-D structure.

The capacitor sits on the source (storage node); the plate is held at $V_{DD}/2$. Its shape is yours as long as the naming rules are kept.

Example Cell (2-D cross-section, names used in the file)



Built with the DEVSIM calls you already use: `add_2d_region (hk)`, `add_2d_contact (storage/plate)`, `add_2d_interface (hk-silicon)`.

Capacitor Design Parameters and Ranges

Parameter	Symbol	Meaning	Allowed range
Height	h	vertical extent of the dielectric above the surface	0.2 – 1.5 μm
Dielectric thickness	t_{diel}	storage node $\leftrightarrow$ plate spacing	3 – 10 nm
Shape	—	two slabs, U-cup, trench, plate ... any 2-D shape	free
Material	ϵ_r	SiO_2 3.9 · Al_2O_3 9 · HfO_2 20 · ZrO_2 35 (by name)	table only
Field	$1 \text{ V} / t_{\text{diel}}$	reliability guard	$\leq 4 \text{ MV/cm}$
2-D estimate	C	$\epsilon \cdot \epsilon_0 \cdot L_{\text{facing}} / t$ per μm , $\times W = 0.1 \mu\text{m}$ (e.g. 12.4 fF)	$\leq 20 \text{ fF (dQ/dV)}$

Important Notes

- Names: bulk / oxide / gate_metal fixed; capacitor dielectric regions = any name starting with hk, storage-node faces = names starting with storage, plate faces = names starting with plate (as many as you need). Details and the self-checker on the submission pages.
- C_{STORE} is what the TA extracts from your file (plate dQ/dV at $V_{DD}/2$). Report your own extraction and the method.

Read Check: Δt -stepped Transient (Python loop around your DC solver)

- Init:** $V_{BL} = 1 \text{ V}$; $V_{\text{cell}} = 0 \text{ V}$ ("0") or 2 V ("1"); $WL 2.5 \text{ V}$; $V_B -0.5 \text{ V}$; 125°C .
- Each $\Delta t = 10 \text{ ps}$:** DC solve $\rightarrow I \rightarrow \Delta Q = I \cdot \Delta t \rightarrow V_{\text{cell}} += \Delta Q / C_{\text{STORE}}$, $V_{BL} -= \Delta Q / C_{BL}$ (Every 10 ps, update the source and drain voltages and rerun the DEVSIM DC simulation to obtain the current).
- Stop at 0.5 ns:** $|V_{BL} - 1 \text{ V}| \geq 60 \text{ mV} \rightarrow \text{pass}$. Plot $I(t)$ and $V_{BL}(t)$ in the report.

Retention Check and Design Hints

- Leakage: data "1" $\rightarrow WL 0 \text{ V}$, $BL 0 \text{ V}$, cell 2 V ; data "0" $\rightarrow WL 0 \text{ V}$, $BL 2 \text{ V}$, cell 0 V ($V_B -0.5 \text{ V}$, 125°C) $\rightarrow I_{\text{leak}}(\text{A}/\mu\text{m}) \times W$
- Time stepping: each Δt , DC $\rightarrow I_{\text{leak}} \rightarrow V_{\text{cell}} -= I_{\text{leak}} \cdot W \cdot \Delta t / C_{\text{STORE}}$ (grow Δt so the cell moves a few mV per step). At 64 ms rerun the 0.5 ns read from that V_{cell} .
- t_{ret} = the time at which that read margin drops below 60 mV. $\geq 64 \text{ ms}$ to pass; longer scores higher.

More margin

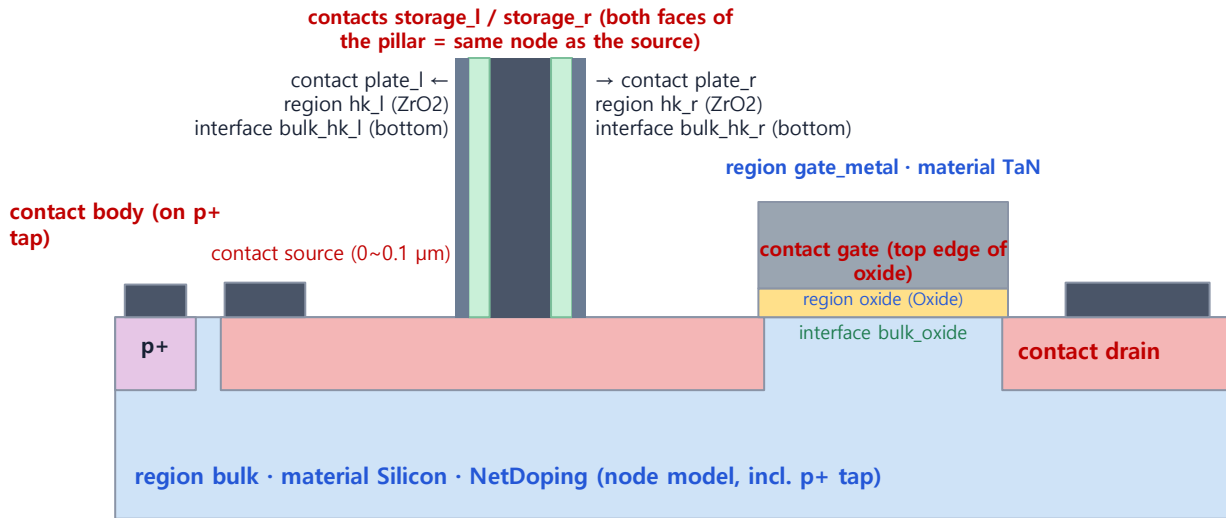
- lower V_{th} , higher ND, shorter S/D $\rightarrow$ more on-current at $V_{GS} 1.5 \text{ V}$

More retention

- the quantity you optimize (scored beyond 64 ms)
- higher V_{th} or bigger C_{STORE} — but watch the "1" read margin

Device design notes: variable names

The TA evaluator opens the submitted file, identifies each part's role from its variable name, and attaches physics and voltages accordingly. Shape, dimensions, mesh and number of contacts are free. Fixed names must match exactly; everything else follows the "starts with ..." rules below, and any structure is then gradable (case-sensitive).



Blue = region · red = contact · green = interface. The figure is an example shape (two slabs). For other shapes (U-cup, trench, horizontal plate, ...) use one dielectric region named hk and as many contacts as needed, e.g. storage_l, storage_r, storage_t. Layout: source on the left, drain on the right, planar gate (the TA assumes this layout when extracting dimensions from the file)

• Also observe

- Materials by name, not number: material of the hk regions (any name starting with hk) = SiO₂-Al₂O₃-HfO₂-ZrO₂, material of gate_metal = a gate name from the table on p.28. bulk is Silicon and oxide is Oxide (SiO₂), fixed
- contacts whose name starts with storage are the storage node (= source potential); contacts starting with plate are at V_{DD}/2. Both must sit on hk regions, and C_STORE is dQ/dV of all plate contacts together
- Regions that need no equations, such as metal pillars or fill space, get a name starting with air, metal or label (the TA physics leaves them alone). Interfaces only between bulk, oxide and hk regions
- A Part 1 device only needs the 'fixed' items (nothing cap-related)

• If it does not converge, use 128-bit extended precision

Type	Variable name	Meaning	Rule
region	bulk	Silicon substrate (channel, n+ S/D and p+ tap all in one region, material Silicon)	fixed
region	oxide	Gate oxide (material Oxide)	fixed
region	gate_metal	Gate-material label region, no equations (material = gate name)	fixed
region	hk, hk_l, hk_r, ... (any name starting with hk)	Storage-cap dielectric. Any number and shape (material = dielectric name)	starts with hk
region	air..., metal..., label... (names starting with these)	Equation-free regions such as fill space or metal pillars (material free)	starts with
contact	gate · source · drain · body	Electrodes where voltages are applied (the TA applies WL, BL and V_B here)	fixed
contact	storage, storage_l, storage_t, ...	Storage-node side of the cap (same node as the source). Must sit on an hk region	starts with storage
contact	plate, plate_l, plate_t, ...	Plate side of the cap (V _{DD} /2). Must sit on an hk region. dQ/dV → C_STORE	starts with plate
interface	bulk_oxide	Silicon-gate oxide boundary (potential continuity)	fixed
interface	(name free) e.g. bulk_hk_l	One wherever bulk, oxide or hk regions touch. A potential-continuity condition is applied	free
node model	NetDoping	Doping of bulk (n-type +, p-type -). The TA physics reads it by this name	fixed

Save the structure — mesh, regions, contacts, interfaces, doping — before any physics is registered.

This file is the design the TA re-runs. Physics equations, biases and results must not be in it.

1. Where in your HW#1 code

```
def build(self):
    self._clear_session()
    self._build_mesh()
    self._build_doping()
    devsim.write_devices(file="part2_[studentID].devsim",
                        device=self.name, type="devsim") # <- add this line
    self._build_physics()
```

- **One line, between doping and physics.** Everything before it (mesh, regions, contacts, interfaces, doping) goes into the file; everything after it does not.
- `device=self.name` is the device name your simulator uses ("mos_light" in the HW#1 code). The simulation continues normally after the save.
- Part 1: save `part1_[studentID].devsim` the same way. One device per file.

2. What must be inside

- **Regions:** bulk (Silicon), oxide (Oxide), gate_metal (gate name from the table on p.42); Part 2 adds the capacitor dielectric regions (names starting with hk, material = dielectric name).
- **Contacts:** gate, source, drain, body; Part 2 adds storage... and plate... contacts on the hk regions.
- **Interfaces:** bulk_oxide, plus one wherever silicon touches an hk region.
- **Doping:** node model NetDoping in bulk.
- **Materials are written as names, never numbers.** Full naming rules: previous slide.
- **If you save after physics registration, the file also contains equations and solution variables — the checker REJECTs it and the TA cannot use it.**

3. Self-check before you submit

- Download `check_structure_file.py` from LearnUs and put it in your step2 folder (next to `mosfet.py`).
- Open an Anaconda Prompt in that folder and run:
 - `..\conda\python.exe check_structure_file.py part2_[studentID].devsim`
 - **OK:** it lists your regions, materials and contacts → take a screenshot for the report.
 - **REJECT:** one reason per line — saved after physics / material name / gate_metal missing / variable name / contact on the wrong region. Fix, save again, re-run.
 - Examples of both on the next slide.

Important Notes

- File names exactly: `part1_[studentID].devsim` and `part2_[studentID].devsim`. No config files, no code are needed for the re-run — the file carries the whole structure.
- The TA also compares the submitted structures with each other; identical files are easy to spot.

Self-check examples before submission

```
Anaconda Prompt
(base) C:\Users\ijun>cd C:\DEVSIM_PROJECT_build\step2
run the self-check
(base) C:\DEVSIM_PROJECT_build\step2>.\conda\python.exe check_structure_file.py samples\ok_cell.devsim
Adding 3673 nodes
```

python executable self-checker devsim file

```
OK: samples\ok_part1_planar.devsim
regions/materials: {'air': 'Air', 'bulk': 'Silicon', 'gate_metal': 'TaN', 'oxide': 'Oxide'}
contacts: ['body', 'drain', 'gate', 'source']
interfaces: ['bulk_oxide']
```

Successful case

```
REJECT: samples\reject_saved_after_physics.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- region 'bulk' 에 방정식 ['PotentialEquation'] 가 들어 있음
- region 'bulk' 에 구조 외 노드 모델 ['IntrinsicCharge', 'IntrinsicCharge:Potential', 'IntrinsicElectrons', 'IntrinsicElectrons:Potential', 'IntrinsicHoles', 'IntrinsicHoles:Potential', 'Potential', 'PotentialIntrinsicCharge'] ...
- region 'oxide' 에 방정식 ['PotentialEquation'] 가 들어 있음
- region 'oxide' 에 구조 외 노드 모델 ['Potential', 'gatenodemodel', 'gatenodemodel:Potential']
- gate_metal 영역이 없어 게이트 재료를 알 수 없습니다: 산화막 위에 방정식 없는 영역 gate_metal 을 만들고 material 에 과제 PPT 재료 표의 게이트 이름을 쓰세요
```

Fail: saved after adding physics equations

```
REJECT: samples\reject_material_numbers.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- 유전막(hk_l) material 에 숫자 '35' 를 썼습니다. 값이 같은 재료 이름 'ZrO2' 로 바꿔 주세요 (이름으로만 받습니다)
- 유전막(hk_r) material 'ZrO2' 는 표에 없습니다. 혹시 ['ZrO2'] 중 하나? 허용 이름: ['SiO2', 'Al2O3', 'HfO2', 'ZrO2']
- 게이트(gate_metal) material 에 숫자 '4.6' 를 썼습니다. 값이 같은 재료 이름 'W' 로 바꿔 주세요 (이름으로만 받습니다)
```

Fail: wrong dielectric name

```
REJECT: samples\reject_variable_names.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- 영역 'HK_R' 는 규칙에 없는 변수명입니다 → 변수명을 'hk_r' 로 바꾸세요
- 접점 'Gate' 는 규칙에 없는 변수명입니다 → 변수명을 'gate' 로 바꾸세요
- 접점 'drain_contact' 는 규칙에 없는 변수명입니다 → 변수명을 'drain' 로 바꾸세요
- 접점 누락: ['drain', 'gate']
- 접점 'plate_r' 는 hk 영역 위에 있어야 하는데 영역 ['HK_R'] 에 있음
- 접점 'storage_r' 는 hk 영역 위에 있어야 하는데 영역 ['HK_R'] 에 있음
- 계면 누락: bulk_oxide (bulk-oxide 사이 계면 'si_ox' 의 이름을 bulk_oxide 로 바꾸세요)
- 계면 'bulk_hk_r' 가 방정식 없는 영역 ['HK_R'] 에 걸려 있음 (계면은 bulk/oxide/hk* 사이에만)
```

Fail: wrong variable names — see p.36

```
REJECT: samples\reject_cap_and_oxide.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- 캡 유전막 ['hk_l', 'hk_r'] 이 있으면 storage* 접점과 plate* 접점이 각각 1개 이상 필요 (현재 storage [], plate ['plate_l', 'plate_r'])
- oxide(게이트 산화막) material 은 Oxide 또는 SiO2 여야 합니다 (현재 'HfO2'). 게이트 산화막은 SiO2 고정
```

Fail: gate oxide is fixed to SiO2

Three things: a PDF report, two device files, and a screenshot of the self-checker saying OK.

Upload to LearnUs by Oct 7, 2026, 11:59 PM. Late submissions get 0 points.

1. Device files (.devsim)

- `part1_[studentID].devsim` — your Part 1 transistor
- `part2_[studentID].devsim` — your Part 2 cell (transistor + capacitor)
- **How to make one:**
 - In your simulator, right after building mesh, regions, contacts, interfaces and doping — **before** registering any equations:
 - ```
devsim.write_devices(file="part2_[studentID].devsim", device=name, type="devsim")
```
  - Use the fixed names and write materials as names (slides 42–46).
  - The TA loads these files and re-runs them. They are the design; the report explains it.

### 2. Report (PDF, about 12 pages, free format)

| Part | Section             | What it must show                                                                                                                        |
|------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | (a) Missing physics | which physics the HW#1 code lacked, why it is needed, how you added and verified it                                                      |
| 1    | (b) Design search   | initial design → what the spec checks showed → how you redesigned (trial → metric → decision); final device figure; table of all 8 specs |
| 1    | (c) Why it works    | change one knob at a time and show which metric moves                                                                                    |
| 2    | (a) Cell structure  | figure of your 1T1C structure; $C_{\text{STORE}}$ from plate dQ/dV                                                                       |
| 2    | (b) Read            | $\Delta V_{\text{BL}}(t)$ curves for “0” and “1”; value at 0.5 ns vs 60 mV                                                               |
| 2    | (c) Design search   | initial design → what the read/retention checks showed → how you redesigned (trial → metric → decision)                                  |
| 2    | (d) Result          | final cell figure; table of read margins and retention time; design rationale                                                            |

Label sections exactly like this (Part-1-(a) ...) so the grader can find them.

### 3. Self-check OK

- Run the checker on both files:
  - `..\conda\python.exe check_structure_file.py part2_[studentID].devsim`
- Paste the screen showing **OK** into the report (slide 47 shows examples).
- **If it says REJECT, the TA cannot re-run your device.** Fix it before submitting — the report is still read, but device results are not credited.

### Important Notes

- Grading re-runs your device files with the TA simulator. Report numbers that do not match the re-run within tolerance are not credited, so leave margin.
- Even if a spec or requirement is not met, a sound record of investigation, implementation and design search earns partial credit — the process is graded, not only the final numbers.
- Questions go to the class group chat so everyone sees the answer.

# 프로젝트 Part 1

**주어진 전기적 스펙을 모두 만족하는 nMOS 소자를 DEVSIM으로 설계하라.**  
HW#1에서 만든 시뮬레이터에서 출발해 구조를 바꾸고, 모든 스펙을 자기 시뮬레이션으로 증명한다.

### 전기적 스펙 ( $V_{DD} = 2\text{ V}$ , 단위는 폭 $1\text{ }\mu\text{m}$ 당)

| 항목           | 조건                                                                                | 기준                                        |
|--------------|-----------------------------------------------------------------------------------|-------------------------------------------|
| $V_{th}$     | 정전류 정의: $V_D = 0.05\text{ V}$ 에서 $I_D = 1\text{e-7 A}/\mu\text{m}$                | $0.45 \pm 0.05\text{ V}$                  |
| $I_{on}$     | $V_G = V_D = 2\text{ V}$ , $300\text{ K}$                                         | $\geq 450\text{ }\mu\text{A}/\mu\text{m}$ |
| $I_{off}$    | $V_G = 0$ , $V_D = 2\text{ V}$ , $300\text{ K}$                                   | $\leq 1\text{ pA}/\mu\text{m}$            |
| SS           | $V_D = 0.05\text{ V}$ , $I_D = 1\text{e-10} \sim 1\text{e-8 A}/\mu\text{m}$ 구간 평균 | $\leq 75\text{ mV/dec}$                   |
| 고온 $I_{off}$ | $V_G = 0$ , $V_D = 2\text{ V}$ , $125\text{ }^\circ\text{C}$ ( $398\text{ K}$ )   | $\leq 100\text{ pA}/\mu\text{m}$          |
| DIBL         | $(V_D = 0.05\text{ V의 } V_{th} - V_D = 2\text{ V의 } V_{th}) / 1.95\text{ V}$      | $\leq 30\text{ mV/V}$                     |
| Body effect  | $V_B = -0.5\text{ V}$ 일 때 $V_{th}$ 상승량 ( $V_D = 0.05\text{ V}$ )                  | $\leq 0.08\text{ V}$                      |
| 산화막 전계       | $E_{ox} = V_{DD} / t_{ox}$                                                        | $\leq 5\text{ MV/cm}$                     |

### 이 표가 뜻하는 것

- 8개 항목은 모두 같은 소자에서 잰다. 구조 하나, 도핑 프로파일 하나, 게이트 재료 하나.
- 일부 항목은 배포된 HW#1 코드로는 측정이 안 된다. 어떤 물리가 빠졌는지 찾아 시뮬레이터에 넣고, 숫자를 믿기 전에 구현을 검증할 것.
- 조건들은 일부러 서로 부딪히게 잡았다.  $I_{on}$ 에 좋은 것은  $I_{off}$ ,  $E_{ox}$ , DIBL에 나쁘다. 해는 존재하며, 균형점을 찾는 것이 과제다.
- 모든 수치는 위의 바이어스·온도·정의 그대로,  $\text{A}/\mu\text{m}$  단위(2D 결과는 폭  $1\text{ }\mu\text{m}$ 당)로 보고한다.

### 주의 사항

- Part 1 제출물: 소자 파일 part1\_[학번].devsim(구조·도핑·재료) + 보고서. 조교가 조교 물리로 이 파일을 재실행하며, 재실행 결과가 합격 여부를 정하므로 여유를 두고 설계할 것.
- Part 1 고정 조건:  $L_g \geq 0.3\text{ }\mu\text{m}$ , 게이트 절연막 =  $\text{SiO}_2$ , 항목에 명시가 없으면  $T = 300\text{ K}$ , 코드 상수(이동도·램프) 임의 수정 금지. 상수를 물리 모델로 대체하는 것은 허용.

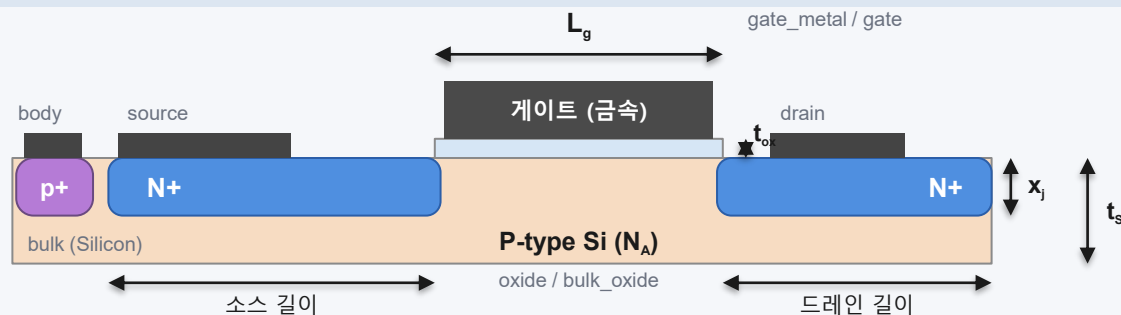
## 소자 치수, 도핑 프로파일, 게이트 재료를 바꿀 수 있다.

이 설계 노브들로 전기적 스펙을 모두 만족시켜라.

### 설계 변수와 허용 범위

| 변수          | 기호       | 무엇을 바꾸나 / 어디에 영향                                   | 허용 범위                       |
|-------------|----------|----------------------------------------------------|-----------------------------|
| 게이트 길이      | $L_g$    | 길게 또는 짧게 ( $V_{th}$ , $I_{on}$ , $I_{off}$ , DIBL) | $\geq 0.3 \mu m$            |
| 게이트 산화막 두께  | $t_{ox}$ | 두껍게 또는 얇게 ( $V_{th}$ , $I_{on}$ , $E_{ox}$ )       | $\geq 4 nm$                 |
| 접합 깊이       | $x_j$    | S/D 접합을 깊게 또는 얇게 (DIBL, 누설)                        | $0.02 - 0.25 \mu m$         |
| 실리콘 두께      | $t_{Si}$ | 시뮬레이션 영역의 기판 깊이                                    | $0.3 - 2 \mu m$             |
| 소스/드레인 길이   | $L_{SD}$ | S/D의 가로 길이 (직렬 저항)                                 | 각 $0.2 - 1 \mu m$           |
| 기판 도핑 (p)   | $N_A$    | 농도와 공간 프로파일                                        | $10^{14} - 10^{17} cm^{-3}$ |
| S/D 도핑 (n+) | $N_D$    | 농도와 공간 프로파일                                        | $10^{18} - 10^{21} cm^{-3}$ |

### 설계 노브 위치 (2D 단면, 파일에 쓰는 변수명 포함)



### 게이트 재료 선택지 ( $SiO_2$ 위 유효 일함수)

| 게이트        | $\Phi_m$ [eV] | 게이트 | $\Phi_m$ [eV] | 게이트        | $\Phi_m$ [eV] |
|------------|---------------|-----|---------------|------------|---------------|
| n+ poly-Si | 4.05          | TaN | 4.45          | Ni         | 5.10          |
| Al         | 4.10          | W   | 4.60          | p+ poly-Si | 5.15          |
| Ta         | 4.25          | TiN | 4.65          | Pt         | 5.30          |
| Ti         | 4.33          | Mo  | 4.70          |            |               |

일함수가 크면  $V_{th}$ 가 오르고 작으면 내려간다. 파일의 gate\_metal 영역에 재료 이름을 쓰면 조교가 표 값을 적용한다. HW#1 코드가 게이트를 무엇으로 가정하고 있는지 확인할 것.

### 설계 힌트 (일반 소자 물리만)

#### $V_{th}$ 를 올리려면

- 일함수가 큰 금속
- 두꺼운  $t_{ox}$ , 높은  $N_A$

#### $I_{on}$ 을 올리려면

- 짧은  $L_g$  ( $\geq 0.3 \mu m$ ), 얇은  $t_{ox}$  ( $\geq 4 nm$ )
- 높은  $N_D$ , 짧은 S/D (직렬 저항 감소)

#### $I_{off}$ 를 줄이려면

- 높은  $V_{th}$  (금속,  $t_{ox}$ ,  $N_A$ )
- 125 °C에서 어떻게 되는지 볼 것

#### DIBL을 줄이려면

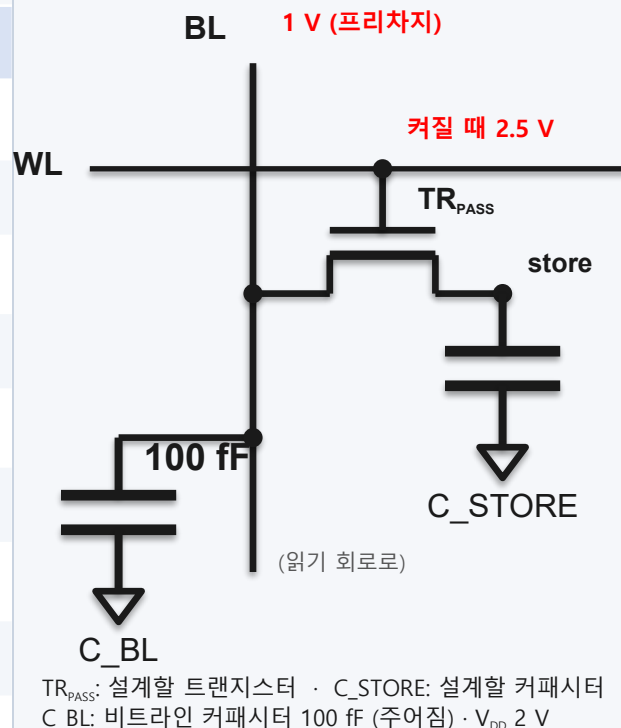
- 긴  $L_g$ , 얇은  $x_j$ , 채널 근처의 높은  $N_A$

### 주의 사항

- 구조·메시·도핑 프로파일을 바꿔도 된다. 숫자 몇 개만 바꾸는 것이 아니다. 필수 영역/접점/계면 변수명은 유지할 것(제출 슬라이드 참조).
- 게이트 절연막은  $SiO_2$ 로 두고 위의 일함수 표를 쓴다. 이동도나 수치 설정을 임의로 바꾸지 말 것. 상수를 물리 모델로 대체하는 것은 허용.

두 소자를 하나의 DEVSIM 구조로 만든다. 읽기와 리텐션 모두  $\Delta t$  스텝 시간축(transient) 시뮬레이션으로 판정한다.

| 항목                             | 값                                        | 비고                                |
|--------------------------------|------------------------------------------|-----------------------------------|
| $V_{DD}$ / BL 프리차지             | 2 V / 1 V                                | 읽기 전마다 BL을 $V_{DD}/2$ 로 맞춤 (프리차지) |
| WL high ( $V_{PP}$ )           | 2.5 V                                    | 읽기 때 pass TR의 게이트 전압              |
| $C_{BL}$                       | 100 fF                                   | 이상적인 비트라인 커패시터                    |
| $C_{STORE}$                    | 학생 설계, 최대 20 fF                          | 자기 커패시터 구조에서 추출 (plate dQ/dV)     |
| 읽기에 필요한 최소 BL 변화, $V_{offset}$ | 60 mV                                    | WL 상승 후 0.5 ns(판정 시점)에 도달해야 함     |
| 저장 상태 초기값                      | "1" = 2 V, "0" = 0 V                     | 쓰기 시뮬레이션 없음. 읽기-리텐션은 이 초기값에서 시작   |
| 온도 / 바디 바이어스                   | 125 °C / $V_b = -0.5$ V                  | Part 2의 모든 시뮬레이션                  |
| Pass TR 치수                     | $L_g = 0.3 \mu m$ 고정, $W = 100$ nm       | 셀 면적 때문. 나머지 노브는 Part 1 범위 안에서 자유 |
| 산화막 전계                         | $E_{ox} = 2.5$ V / $t_{ox} \leq 5$ MV/cm | 게이트에 WL 2.5 V가 걸림                 |



WL이 켜진 뒤  $t = 0.5 \text{ ns}$ (판정 시점)에  $|V_{BL} - 1 \text{ V}| \geq 60 \text{ mV}$ , 두 저장 상태 모두 자기 DC 소자로  $\Delta t$  스텝 transient를 돌려 계산한다(배경 참조).

WL 0, 125 °C,  $V_R$  -0.5 V에서 수행.

데이터 "0" 기준:  $V_{BL}$  2V, 데이터 "1" 기준  $V_{BL}$  0V로 설정 후 수행.

셀 전압을  $\Delta t$  스텝 시간축 시뮬레이션으로 따라가  
고, 64 ms 뒤 같은 읽기 transient에서 마진 60  
mV 이상이어야 함. 리텐션 시간 = 마진이 60 mV  
아래로 내려가는 시각. 길수록 좋다.

게이트에 2.5 V가 걸릴 때  $E_{ox} \leq 5 \text{ MV/cm} \rightarrow t_{ox} \geq 5 \text{ nm}$ . 커패시터 유전막 전계  $1 \text{ V} / t_{diel} \leq 4 \text{ MV/cm}$ .

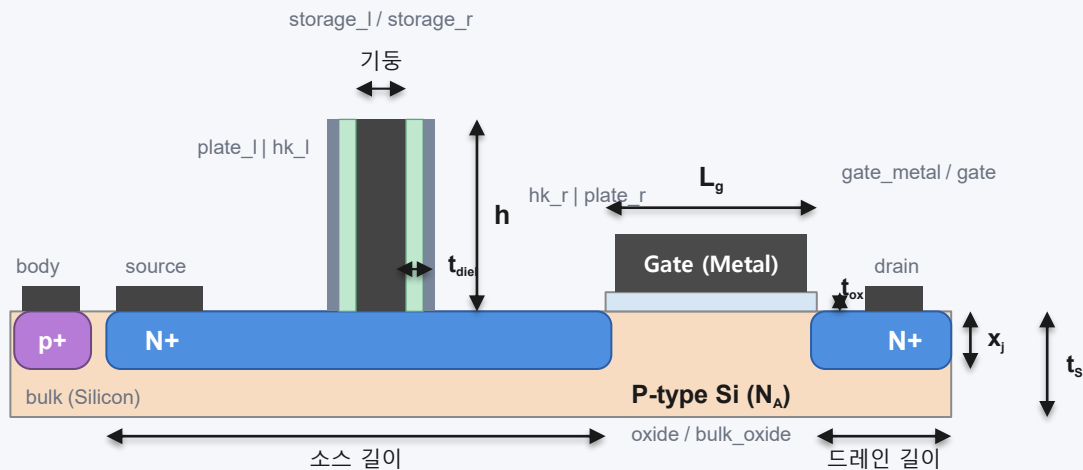
왜 어려운가:  $V_{th}$ 가 낮으면 빨리 읽히지만 샌다. 높으면 데이터는 지키지만 0.5 ns 안에 옮기는 전하가 줄어 "1" 읽기 마진이 얇아진다. Part 1 레퍼런스는 125 °C 리텐션에 떨어진다. 게이트 금속·도핑·커패시터를 함께 조절할 것.

- Part 2 제출물: pass 트랜지스터와 커패시터가 한 구조에 든 part2\_[학번].devsim + 읽기 transient 곡선, 리텐션 추정,  $C_{STORE}$  추출을 담은 보고서.
- 조교가 파일을 열어 조교 물리를 엮고 같은 읽기/리텐션 절차를 반복한다. 조교 허용 오차 안의 결과만 인정되니 여유를 둘 것.

## pass 트랜지스터와 저장 커패시터를 하나의 2D 구조로 설계한다.

커패시터는 소스(저장 노드) 위에 얹고 플레이트는  $V_{DD}/2$ 에 둔다. 변수명 규칙만 지키면 모양은 자유다.

### 예시 셀 (2D 단면, 파일에 쓰는 변수명 포함)



이미 쓰고 있는 DEVSIM 호출로 만든다: add\_2d\_region(hk), add\_2d\_contact(storage/plate), add\_2d\_interface(hk-실리콘).

### 커패시터 설계 변수와 범위

| 변수     | 기호                              | 뜻                                                                                                                    | 허용 범위                        |
|--------|---------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------|
| 높이     | $h$                             | 표면 위 유전막의 세로 길이                                                                                                      | 0.2 – 1.5 $\mu\text{m}$      |
| 유전막 두께 | $t_{\text{diel}}$               | 저장 노드와 플레이트 사이 간격                                                                                                    | 3 – 10 nm                    |
| 모양     | —                               | 두 슬랩, U자 컵, 트렌치, 판 ... 어떤 2D 모양이든                                                                                    | 자유                           |
| 재료     | $\epsilon_r$                    | SiO <sub>2</sub> 3.9 · Al <sub>2</sub> O <sub>3</sub> 9 · HfO <sub>2</sub> 20 · ZrO <sub>2</sub> 35 (이름으로)           | 표의 값만                        |
| 전계     | $1 \text{ V} / t_{\text{diel}}$ | 신뢰성 한계                                                                                                               | $\leq 4 \text{ MV/cm}$       |
| 2D 추정  | $C$                             | $\epsilon \cdot \epsilon_0 \cdot \text{마주보는 길이} / t \text{ (}\mu\text{m당)}, \times W = 0.1 \mu\text{m}$ (예: 12.4 fF) | $\leq 20 \text{ fF (dQ/dV)}$ |

### 주의 사항

- 변수명: bulk / oxide / gate\_metal은 고정; 커패시터 유전막 영역은 hk로 시작하는 이름, 저장 노드 면은 storage로 시작하는 이름, 플레이트 면은 plate로 시작하는 이름 (개수는 필요한 만큼). 세부 규칙과 자가 검사는 제출 슬라이드 참조.
- $C_{\text{STORE}}$ 는 조교가 파일에서 추출한 값( $V_{DD}/2$ 에서 plate dQ/dV)을 쓴다. 자기 추출값과 방법도 보고할 것.

### 읽기 판정: $\Delta t$ 스텝 transient (DEVSIM를 파이썬 루프로 감싼다)

- 초기값:  $V_{BL} = 1 \text{ V}$ ;  $V_{\text{cell}} = 0 \text{ V}$  ("0") or  $V_{\text{cell}} = 2 \text{ V}$  ("1");  $W_L 2.5 \text{ V}$ ;  $V_B -0.5 \text{ V}$ ; 125 °C.
- 매  $\Delta t = 10 \text{ ps}$ : DC 해석  $\rightarrow I \rightarrow \Delta Q = I \cdot \Delta t \rightarrow V_{\text{cell}} += \Delta Q / C_{\text{STORE}}$   $V_{BL} -= \Delta Q / C_{BL}$  (10ps 마다 DEVSIM에서의 drain, source voltage update  $\rightarrow$  devsim의 반복으로 측정)
- 0.5 ns에서 멈춤:  $|V_{BL} - 1 \text{ V}| \geq 60 \text{ mV}$ 면 통과.  $I(t)$ 와  $V_{BL}(t)$ 를 보고서에 그릴 것.

### 리텐션 판정과 설계 힌트

- 누설: data "1"  $\rightarrow W_L 0 \text{ V}$ ,  $B_L 0 \text{ V}$ , 셀 2 V; data "0"  $\rightarrow W_L 0 \text{ V}$ ,  $B_L 2 \text{ V}$ , 셀 0 V ( $V_B -0.5 \text{ V}$ , 125 °C)  $\rightarrow I_{\text{leak}}(\text{A}/\mu\text{m}) \times W$
- 시간축 적분: 매  $\Delta t$ 마다 DC  $\rightarrow I_{\text{leak}} \rightarrow V_{\text{cell}} -= I_{\text{leak}} \cdot W \cdot \Delta t / C_{\text{STORE}}$  ( $\Delta t$ 는 스텝당 수 mV 이하로 키워 감). 64 ms 시점의  $V_{\text{cell}}$ 에서 0.5 ns 읽기를 다시 돌린다.
- $t_{\text{ret}}$  = 그 읽기 마진이 60 mV 아래로 내려가는 시각. 64 ms 이상이면 통과, 길수록 점수가 높다.

#### 마진을 키우려면

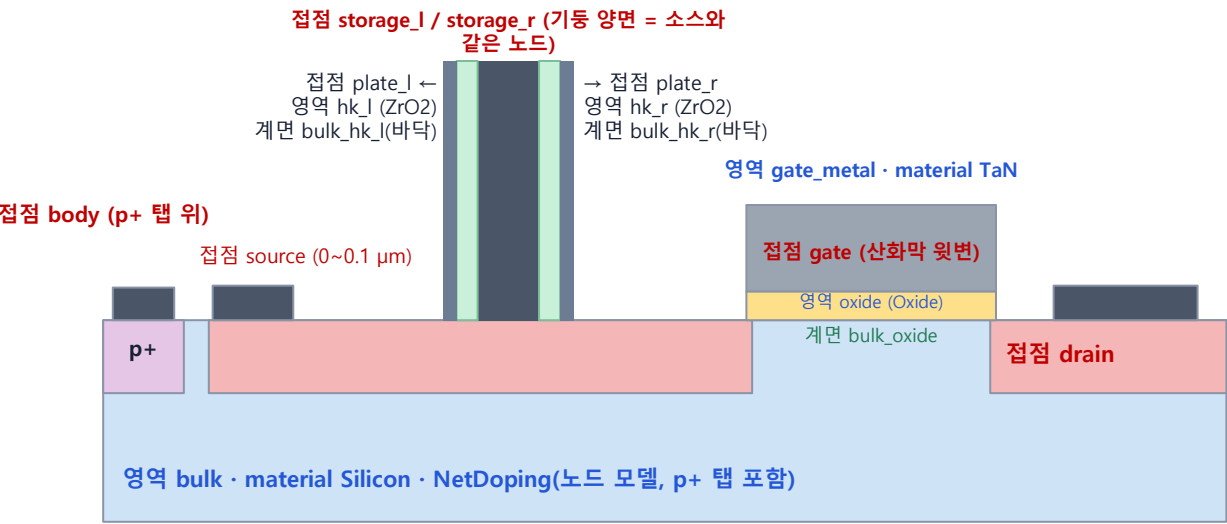
- 낮은  $V_{th}$ , 높은 ND, 짧은 S/D  $\rightarrow V_{GS} 1.5 \text{ V}$ 의 on-current  $\uparrow$

#### 리텐션을 늘리려면

- 최적화 대상 (64 ms 이상은 점수화)
- 높은  $V_{th}$  또는 큰  $C_{\text{STORE}}$  — 단, "1" 읽기 마진(속도)을 확인할 것

# 소자 설계 시 주의점: 변수명

조교 평가기는 제출 파일을 열어 변수명(이름)으로 역할을 알아내 물리식과 전압을 붙인다. 구조 모양·치수·메시·접점 개수는 자유. 고정 이름은 글자 그대로, 나머지는 아래의 '...로 시작하는 이름' 규칙을 따르면 어떤 구조든 채점된다 (대소문자 구분).



| 종류    | 변수명                                   | 뜻                                                 | 규칙          |
|-------|---------------------------------------|---------------------------------------------------|-------------|
| 영역    | bulk                                  | 실리콘 기판 (채널·n+ S/D·p+ 탭 전부 한 영역, material Silicon) | 고정          |
| 영역    | oxide                                 | 게이트 산화막 (material Oxide)                          | 고정          |
| 영역    | gate_metal                            | 게이트 재료 라벨 영역, 방정식 없음 (material = 게이트 이름)          | 고정          |
| 영역    | hk, hk_l, hk_r, ... (hk로 시작하는 이름)     | 저장 캡 유전막. 개수·모양 자유 (material = 유전막 이름)            | hk로 시작      |
| 영역    | air..., metal..., label... (이 단어로 시작) | 채움 공간·금속 기둥 등 방정식 없는 영역 (재료 자유)                   | 이 단어로 시작    |
| 접점    | gate · source · drain · body          | 전압을 거는 전극 (조교가 WL·BL·V_B 를 여기에 인가)                | 고정          |
| 접점    | storage, storage_l, storage_t, ...    | 캡 저장 노드 쪽 면 (소스와 같은 노드). hk 영역 위에                 | storage로 시작 |
| 접점    | plate, plate_l, plate_t, ...          | 캡 플레이트 쪽 면 (V_DD/2). hk 영역 위에 . dQ/dV → C_STORE   | plate로 시작   |
| 계면    | bulk_oxide                            | 실리콘-게이트 산화막 경계 (전위 연속)                            | 고정          |
| 계면    | (이름 자유) 예: bulk_hk_l                  | bulk-oxide-hk 영역끼리 맞닿는 곳마다 하나씩. 전위 연속 조건이 붙음      | 자유          |
| 노드 모델 | NetDoping                             | bulk 의 도핑 값 (n형 +, p형 -). 조교 물리 가 이 이름으로 읽음       | 고정          |

파랑 = 영역(region) · 빨강 = 접점(contact) · 초록 = 계면(interface). 그림은 예시 형태(두 슬랩). U자·트렌치·가로 판 등 다른 모양이면 유전막 영역은 hk 하나, 접점은 storage\_l, storage\_r, storage\_t 처럼 필요한 만큼 만들면 됨. 배치는 소스 왼쪽·드레인 오른쪽·게이트 평탄 (조교가 파일에서 치수를 역추출할 때 이 배치를 가정)

- 함께 지킬 것
  - 재료는 숫자가 아니라 이름: hk 영역(hk로 시작하는 이름)의 material = SiO2·Al2O3·HfO2·ZrO2, gate\_metal 의 material = 50p 표의 게이트 이름. bulk 는 Silicon, oxide 는 Oxide(SiO2) 고정
  - 이름이 storage로 시작하는 접점은 저장 노드(= 소스 전위), plate로 시작하는 접점은 V\_DD/2. 둘 다 hk 영역 위에 있어야 하고, C\_STORE 는 plate 접점 전체의 dQ/dV
  - 금속 기둥·채움 공간처럼 방정식이 필요 없는 영역은 air, metal, label 중 하나로 시작하는 이름으로 (조교 물리가 건드리지 않음). 계면은 bulk-oxide-hk 영역 사이에만
  - Part 1 소자는 '고정' 항목만 있으면 됨 (캡 관련 없음)

- 수렴이 안되면 128비트 확장 정밀도 사용

## 메시·영역·접점·계면·도핑까지의 구조를, 물리를 등록하기 전에 저장한다.

이 파일이 조교가 재실행하는 설계 자체다. 물리식·바이어스·결과가 들어 있으면 안 된다.

### 1. HW#1 코드의 어디에

```
def build(self):
 self._clear_session()
 self._build_mesh()
 self._build_doping()
 devsim.write_devices(file="part2_[studentID].devsim",
 device=self.name, type="devsim") # <- 이 줄을 추가
 self._build_physics()
```

- 도핑과 물리 사이에 한 줄. 그 앞에서 만든 것(메시·영역·접점·계면·도핑)은 파일에 들어가고, 뒤의 것은 들어가지 않는다.
- device=self.name 은 시뮬레이터가 쓰는 소자 이름(HW#1 코드에서는 "mos\_light"). 저장 뒤 시뮬레이션은 평소대로 이어진다.
- Part 1도 같은 방법으로 part1\_[학번].devsim 저장. 파일 하나에 소자 하나.

### 2. 파일에 들어가야 하는 것

- 영역: bulk(Silicon), oxide(Oxide), gate\_metal(51p 표의 게이트 이름); Part 2는 커패시터 유전막 영역 추가(hk로 시작하는 이름, material = 유전막 이름).
- 접점: gate, source, drain, body; Part 2는 hk 영역 위에 storage..., plate... 접점 추가.
- 계면: bulk\_oxide, 그리고 실리콘이 hk 영역에 닿는 곳마다 하나씩.
- 도핑: bulk의 노드 모델 NetDoping.
- 재료는 숫자가 아니라 이름으로 쓴다. 변수명 규칙 전체는 앞장.
- 물리 등록 뒤에 저장하면 방정식과 해 변수까지 파일에 들어간다. 검사기가 REJECT하고 조교도 쓸 수 없다.

### 3. 제출 전 자가 검사

- LearnUs에서 check\_structure\_file.py를 받아 자기 step2 폴더(mosfet.py 옆)에 넣는다.
- 그 폴더에서 Anaconda Prompt를 열고 실행:  
..W\condaWpython.exe check\_structure\_file.py part2\_[학번].devsim
- OK: 영역·재료·접점 목록이 찍힌다 → 화면을 캡처해 보고서에 넣는다.
- REJECT: 이유가 줄마다 나온다 — 물리 등록 뒤 저장 / 재료 이름 / gate\_metal 없음 / 변수명 / 접점이 엉뚱한 영역. 고치고 다시 저장한 뒤 재실행.
- 두 경우의 예시는 다음 장.

### 주의 사항

- 파일 이름은 정확히 part1\_[학번].devsim, part2\_[학번].devsim. 재실행에 config나 코드는 필요 없다. 파일이 구조 전체를 담는다.
- 조교는 제출된 구조끼리도 비교한다. 같은 파일은 바로 드러난다.

# 제출 전 자가 검사 예시

```
Anaconda Prompt
(base) C:\Users\jjun>cd C:\DEVSIM_PROJECT_build\step2
자가검사 실행
(base) C:\DEVSIM_PROJECT_build\step2>.\.conda\python.exe check_structure_file.py samples\ok_cell.devsim
Adding 3673 nodes
```

python 실행 exe 자가검사기

devsim  
파일

```
OK: samples\ok_part1_planar.devsim
regions/materials: {'air': 'Air', 'bulk': 'Silicon', 'gate_metal': 'TaN', 'oxide': 'Oxide'}
contacts: ['body', 'drain', 'gate', 'source']
interfaces: ['bulk_oxide']
```

## 성공 사례

```
REJECT: samples\reject_saved_after_physics.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- region 'bulk' 에 방정식 ['PotentialEquation'] 가 들어 있음
- region 'bulk' 에 구조 외 노드 모델 ['IntrinsicCharge', 'IntrinsicCharge:Potential', 'IntrinsicElectrons', 'IntrinsicElectrons:Potential', 'IntrinsicHoles', 'IntrinsicHoles:Potential', 'Potential', 'PotentialIntrinsicCharge'] ...
- region 'oxide' 에 방정식 ['PotentialEquation'] 가 들어 있음
- region 'oxide' 에 구조 외 노드 모델 ['Potential', 'gatenodemodel', 'gatenodemodel:Potential']
- gate_metal 영역이 없어 게이트 재료를 알 수 없습니다: 산화막 위에 방정식 없는 영역 gate_metal 을 만들고 material 에 과제 PPT 재료 표의 게이트 이름을 쓰세요
```

## 실패: 물리수식 넣은 뒤 저장

```
REJECT: samples\reject_material_numbers.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- 유전막(hk_l) material 에 숫자 '35' 를 썼습니다. 값이 같은 재료 이름 'ZrO2' 로 바꿔 주세요 (이름으로만 받습니다)
- 유전막(hk_r) material 'ZrO2' 는 표에 없습니다. 혹시 ['ZrO2'] 중 하나? 허용 이름: ['SiO2', 'Al2O3', 'HfO2', 'ZrO2']
- 게이트(gate_metal) material 에 숫자 '4.6' 를 썼습니다. 값이 같은 재료 이름 'W' 로 바꿔 주세요 (이름으로만 받습니다)
```

## 실패: 유전체 이름 오류

```
REJECT: samples\reject_variable_names.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- 영역 'HK_R' 는 규칙에 없는 변수명입니다 → 변수명을 'hk_r' 로 바꾸세요
- 접점 'Gate' 는 규칙에 없는 변수명입니다 → 변수명을 'gate' 로 바꾸세요
- 접점 'drain_contact' 는 규칙에 없는 변수명입니다 → 변수명을 'drain' 로 바꾸세요
- 접점 누락: ['drain', 'gate']
- 접점 'plate_r' 는 hk 영역 위에 있어야 하는데 영역 ['HK_R'] 에 있음
- 접점 'storage_r' 는 hk 영역 위에 있어야 하는데 영역 ['HK_R'] 에 있음
- 계면 누락: bulk_oxide (bulk-oxide 사이 계면 'si_ox' 의 이름을 bulk_oxide 로 바꾸세요)
- 계면 'bulk_hk_r' 가 방정식 없는 영역 ['HK_R'] 에 걸려 있음 (계면은 bulk/oxide/hk* 사이에만)
```

## 실패: 변수명 오류 변수명 53p 참고

```
REJECT: samples\reject_cap_and_oxide.devsim
아래 항목을 고친 뒤 다시 저장해 주세요:
- 캡 유전막 ['hk_l', 'hk_r'] 이 있으면 storage* 접점과 plate* 접점이 각각 1개 이상 필요 (현재 storage [], plate ['plate_l', 'plate_r'])
- oxide(게이트 산화막) material 은 Oxide 또는 SiO2 여야 합니다 (현재 'HfO2'). 게이트 산화막은 SiO2 고정
```

## 실패: gate oxide는 SiO2로 고정

## 세 가지: 보고서 PDF, 소자 파일 2개, 자가 검사기 OK 화면 캡처.

2026년 10월 7일 23:59까지 LearnUs에 업로드. 시각 제출은 0점.

## 1. 소자 파일 (.devsim)

- part1\_[학번].devsim — Part 1 트랜지스터
- part2\_[학번].devsim — Part 2 셀 (트랜지스터 + 커패시터)
- 만드는 법:
  - 자기 시뮬레이터에서 메시:영역:접점:계면:도핑을 만든 직후, 방정식을 등록하기 전에:
  - `devsim.write_devices(file="part2_[학번].devsim", device=이름, type="devsim")`
  - 고정 변수명을 지키고 재료는 이름으로 쓴다 (51~55p).
  - 조교는 이 파일을 열어 재실행한다. 파일이 설계이고, 보고서는 그 설명이다.

## 2. 보고서 (PDF, 약 12쪽, 양식 자유)

| Part | 절         | 들어가야 할 내용                                                          |
|------|-----------|--------------------------------------------------------------------|
| 1    | (a) 빠진 물리 | HW#1 코드에 어떤 물리가 없었는지, 왜 필요한지, 어떻게 넣고 검증했는지                         |
| 1    | (b) 설계 탐색 | 처음 설계 → 스펙 검사 결과 → 어떻게 다시 설계했는지 (시도 → 지표 → 판단); 최종 소자 그림; 스펙 8항목 표 |
| 1    | (c) 왜 되는가 | 노브를 하나씩 바꿔 어느 지표가 움직이는지 보일 것                                       |
| 2    | (a) 셀 구조  | 1T1C 구조 그림; plate dQ/dV로 구한 $C_{STORE}$                            |
| 2    | (b) 읽기    | "0"과 "1"의 $\Delta V_{BL}(t)$ 곡선; 0.5 ns 값과 60 mV 비교                |
| 2    | (c) 설계 탐색 | 처음 설계 → 읽기-리텐션 검사 결과 → 어떻게 다시 설계했는지 (시도 → 지표 → 판단)                 |
| 2    | (d) 결과    | 최종 셀 그림; 읽기 마진과 리텐션 시간 표; 설계 근거                                    |

절 이름을 이 형식(Part-1-(a) ...) 그대로 붙여 채점자가 찾기 쉽게 할 것.

## 3. 자가 검사 OK

- 두 파일 모두 검사기를 돌린다:
  - `..W\conda\python.exe check_structure_file.py part2_[학번].devsim`
- OK가 뜬 화면을 보고서에 붙인다 (예시는 56p).
- REJECT가 뜨면 조교가 소자를 재실행할 수 없다. 제출 전에 고칠 것. 보고서는 읽지만 소자 결과는 인정되지 않는다.

## 주의 사항

- 채점은 조교 시뮬레이터로 소자 파일을 재실행해서 한다. 재실행과 허용 오차 밖으로 다른 보고서 수치는 인정되지 않으니 여유를 둘 것.
- 스펙이나 요구사항에 미달해도 조사·구현·설계 탐색 기록이 타당하면 부분 점수를 준다. 최종 수치만이 아니라 과정을 평가한다.
- 질문은 수업 단체 채팅방에 올려 모두가 답을 볼 수 있게 한다.